Supplementary Figures 13-53: CCT-based Quality Control

Performance of Additional Sequencing Models

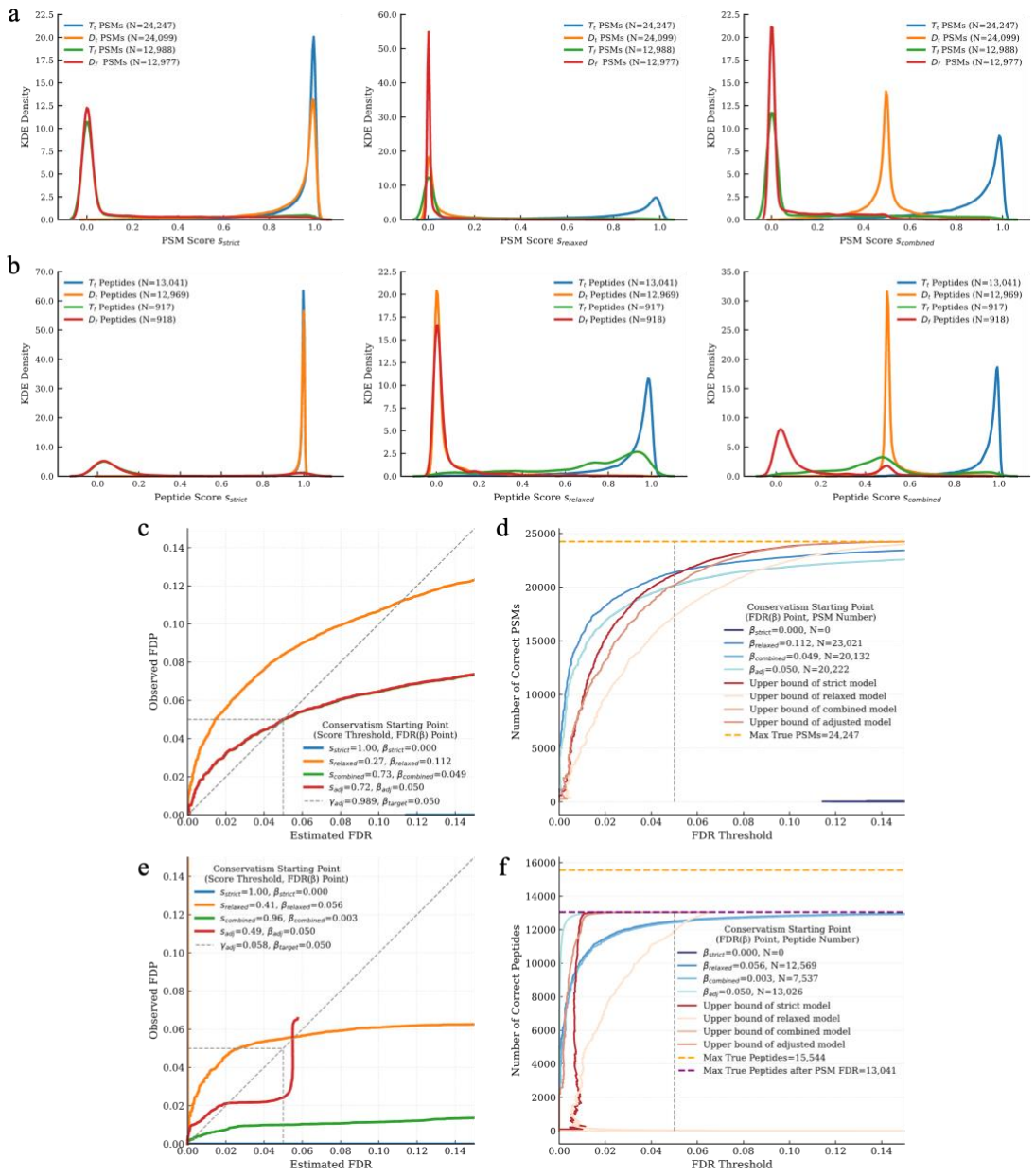


Figure S13: Dual-level CCT QC on GraphNovo test set using π-xNovo-DDA trained on the GraphNovo training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

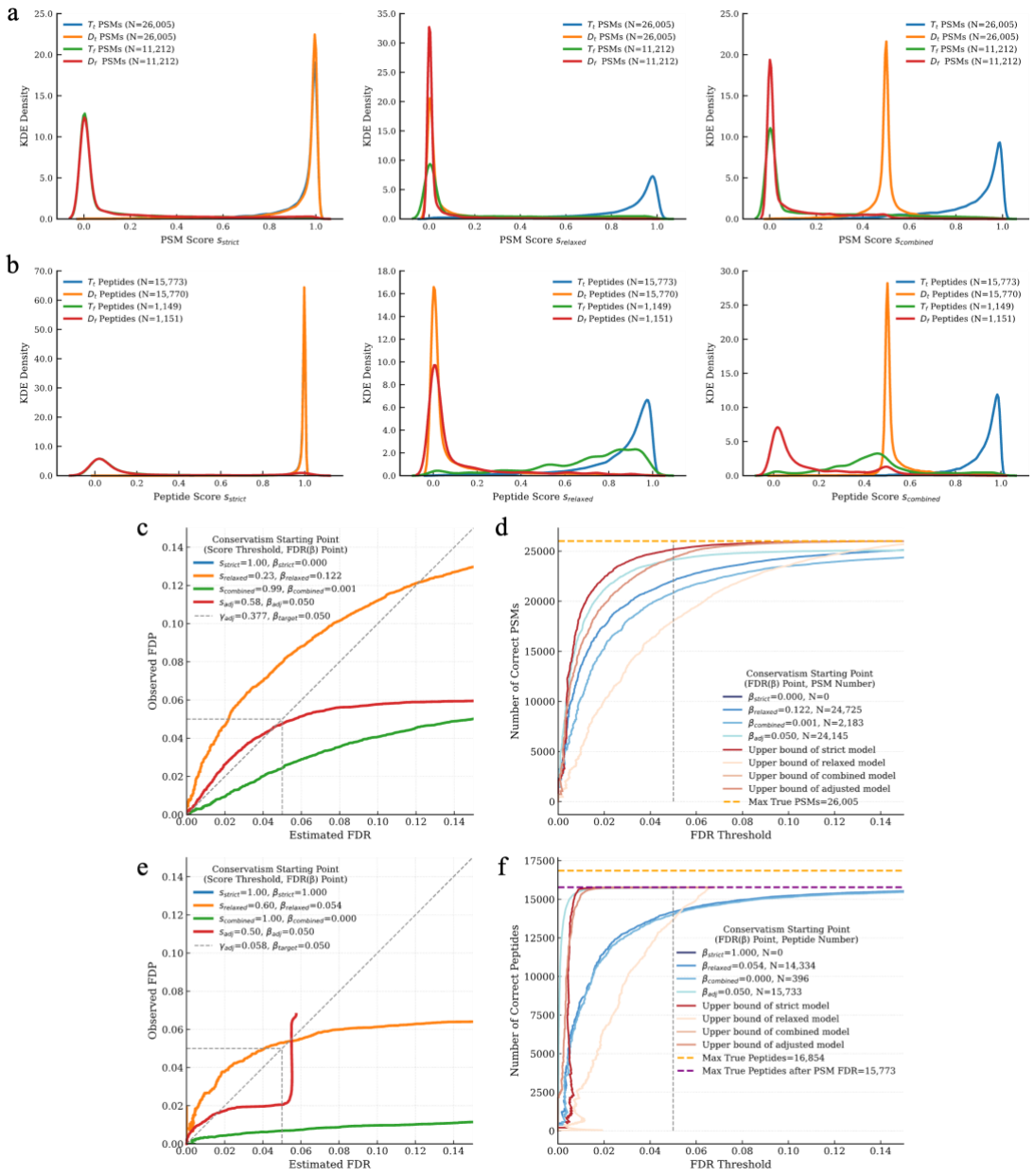


Figure S14: Dual-level CCT QC on GraphNovo test set using π-HelixNovo trained on the MassIVE-KB training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

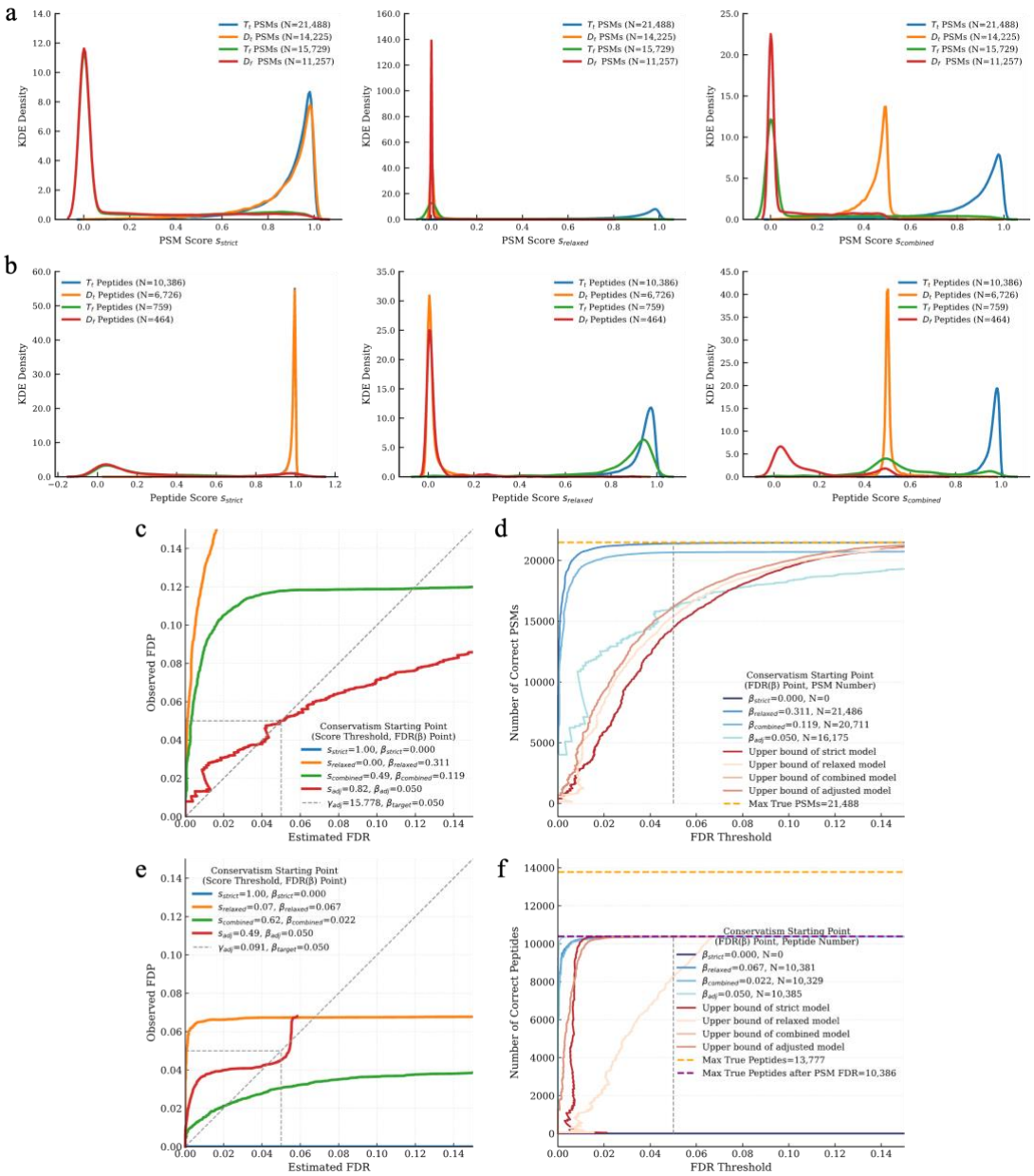


Figure S15: Dual-level CCT QC on GraphNovo test set using π-HelixNovo trained on the GraphNovo training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

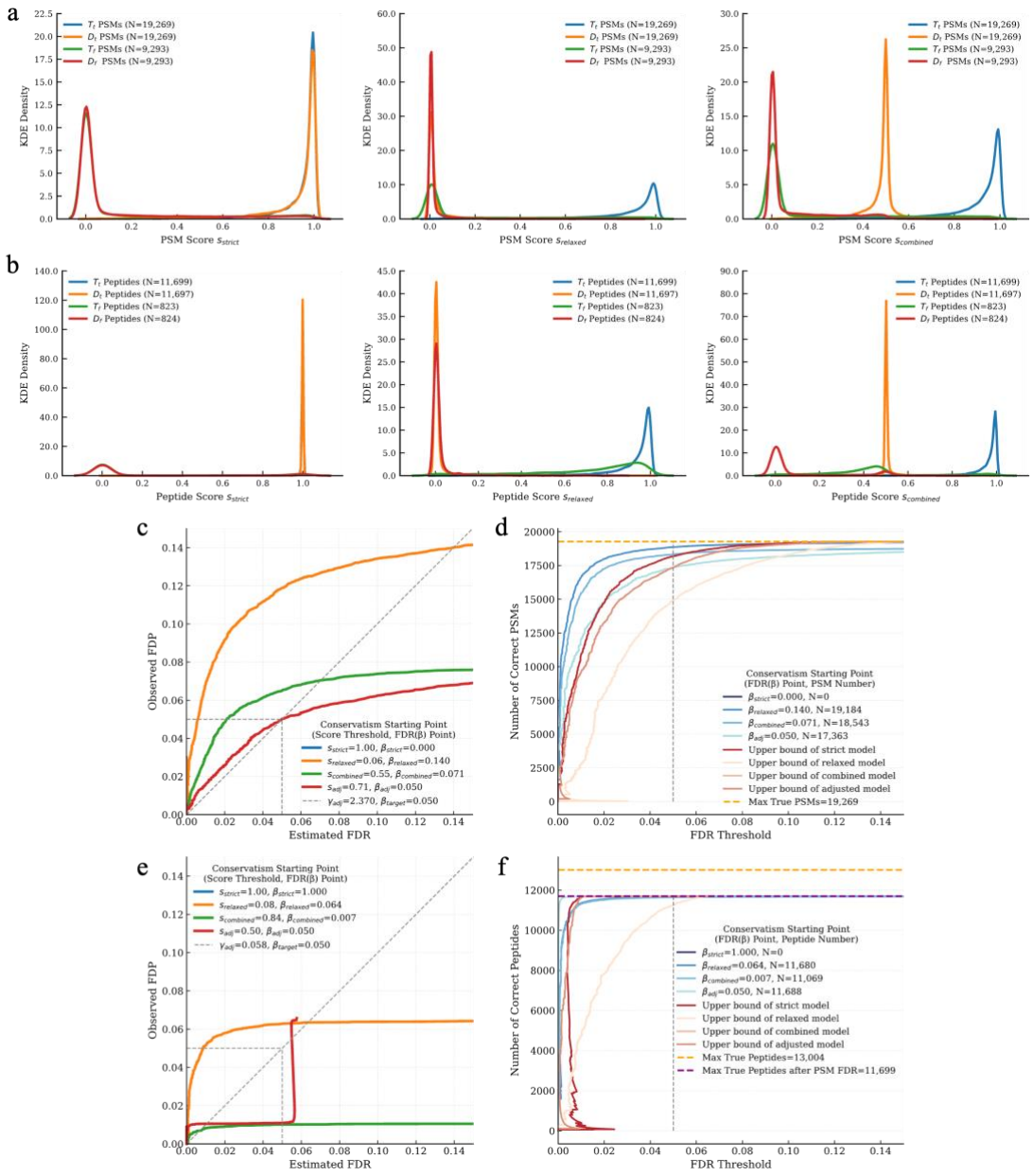


Figure S16: Dual-level CCT QC on GraphNovo test set using Casanovo trained on the MassIVE-KB training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

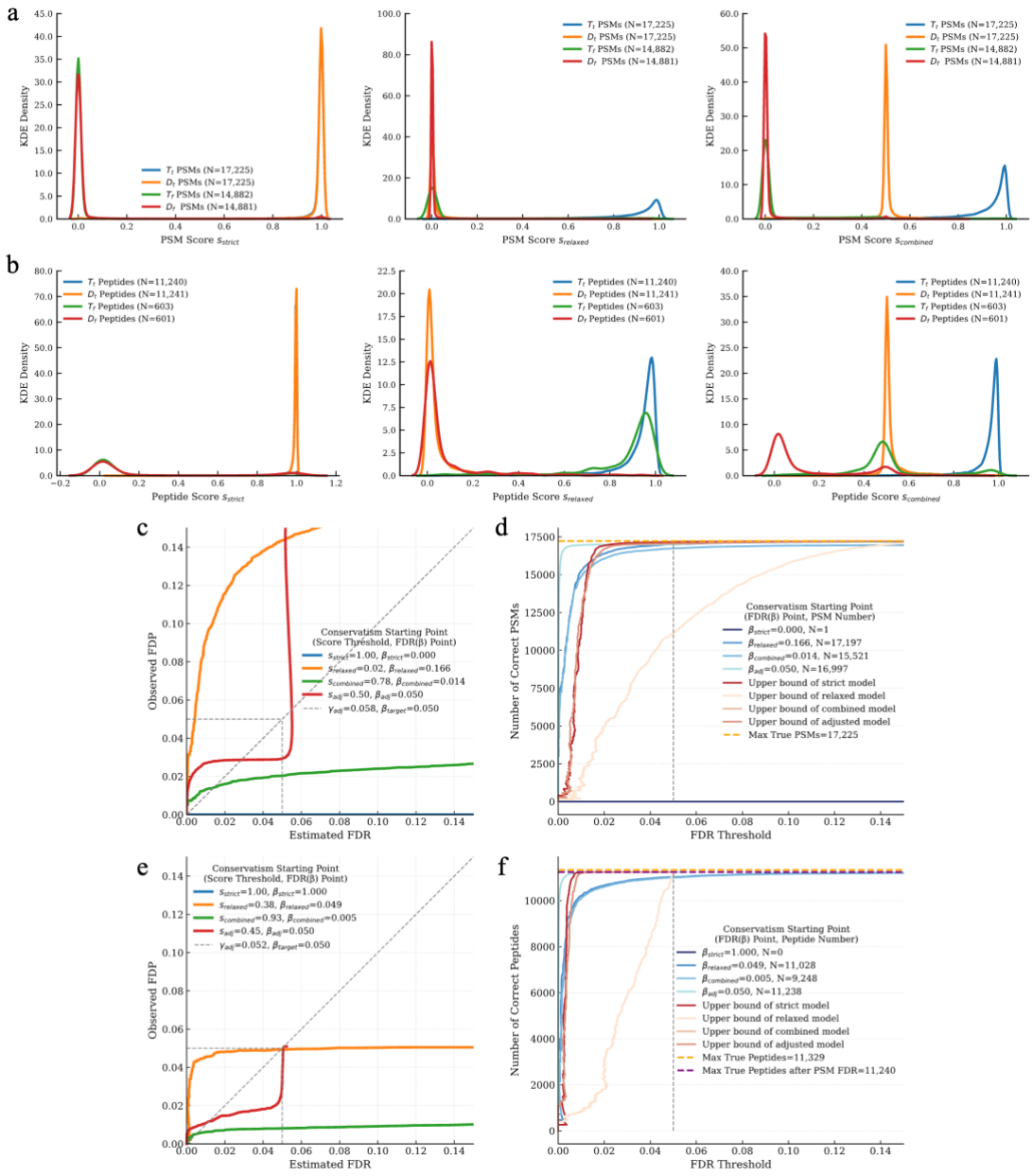


Figure S17: Dual-level CCT QC on GraphNovo test set using Casanovo trained on the GraphNovo training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

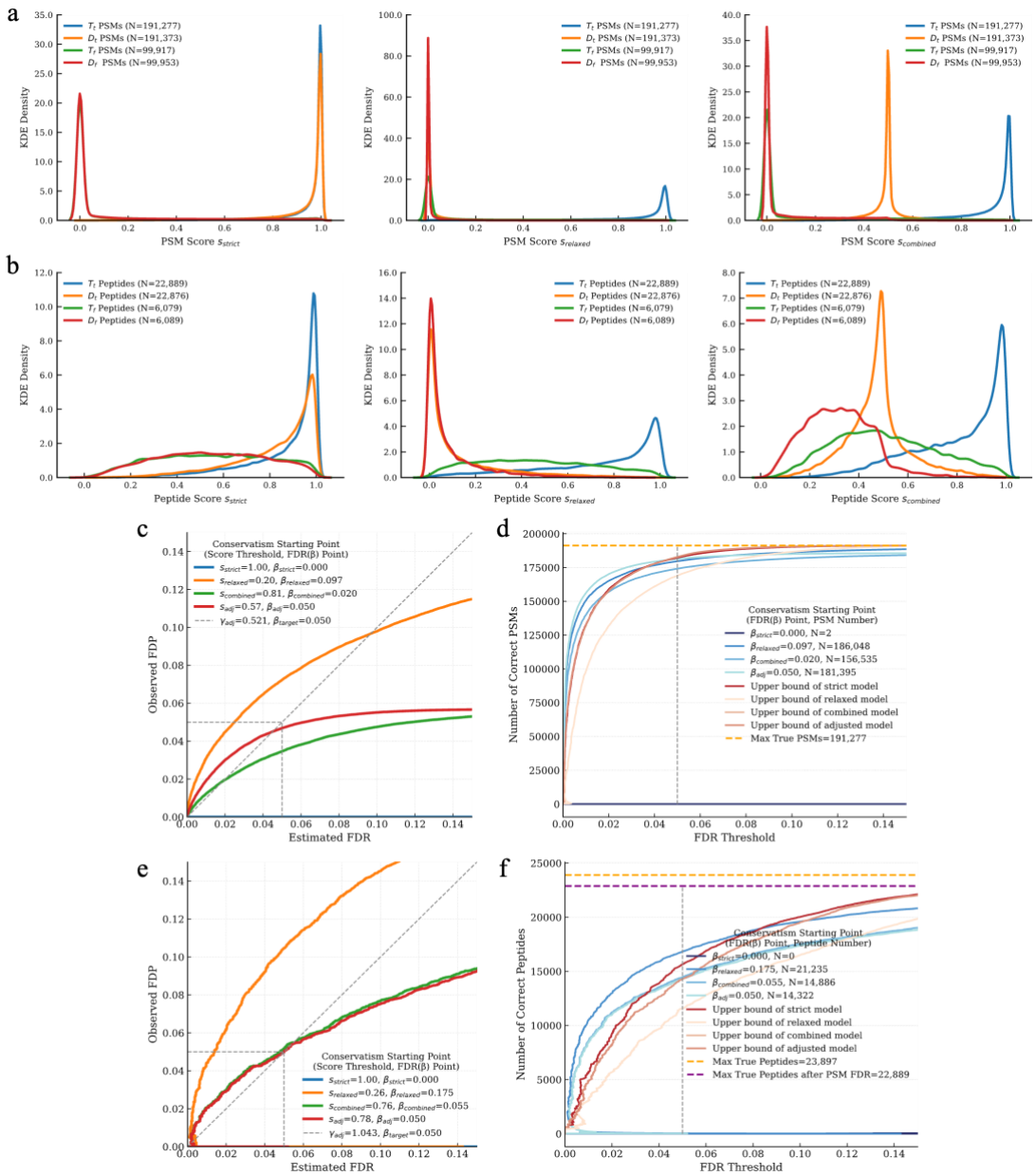


Figure S18: Dual-level CCT QC on *Bacillus* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

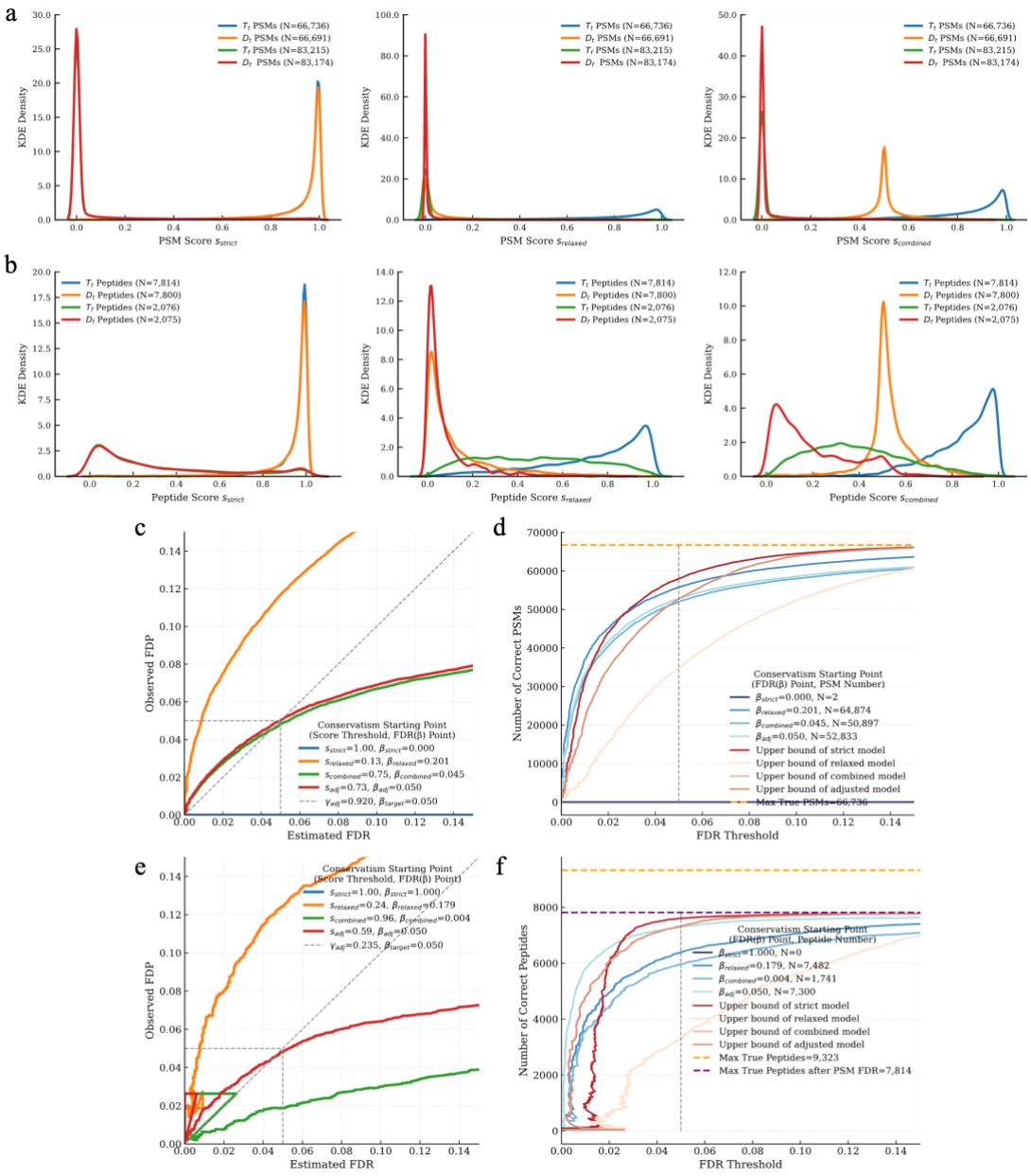


Figure S19: Dual-level CCT QC on *C. endoloripes* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

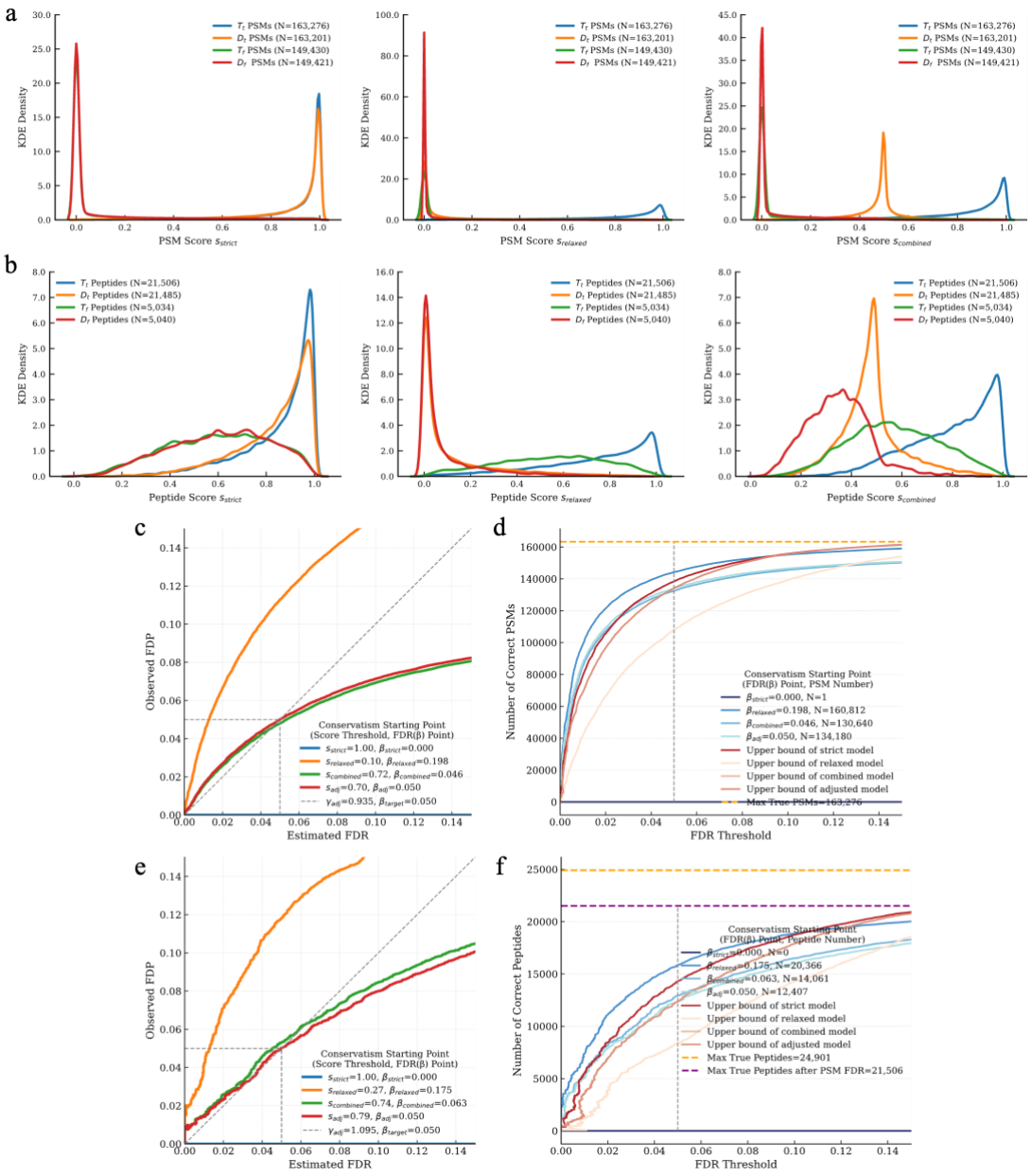


Figure S20: Dual-level CCT QC on *A. mellifera* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

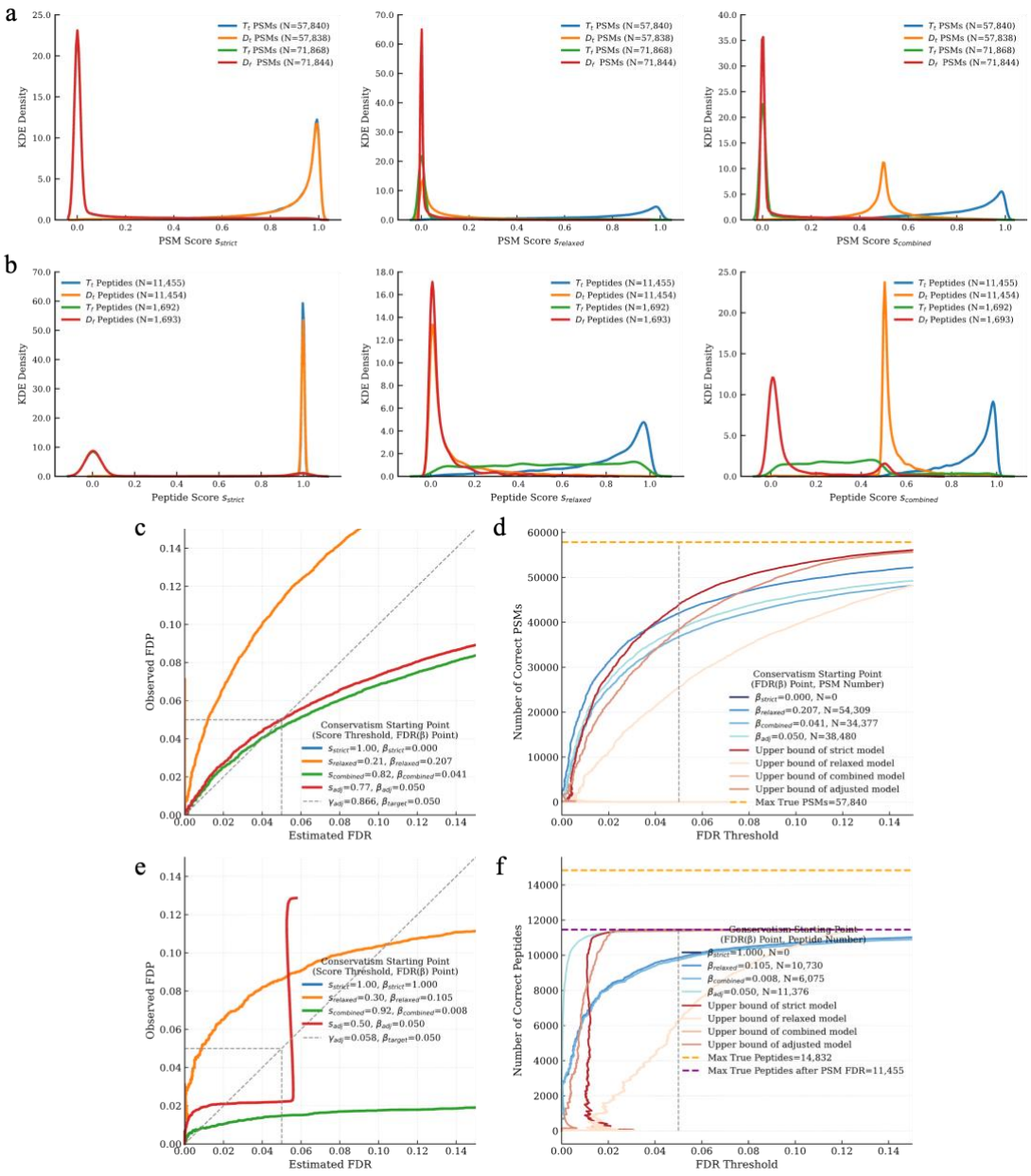


Figure S21: Dual-level CCT QC on *H. sapiens* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

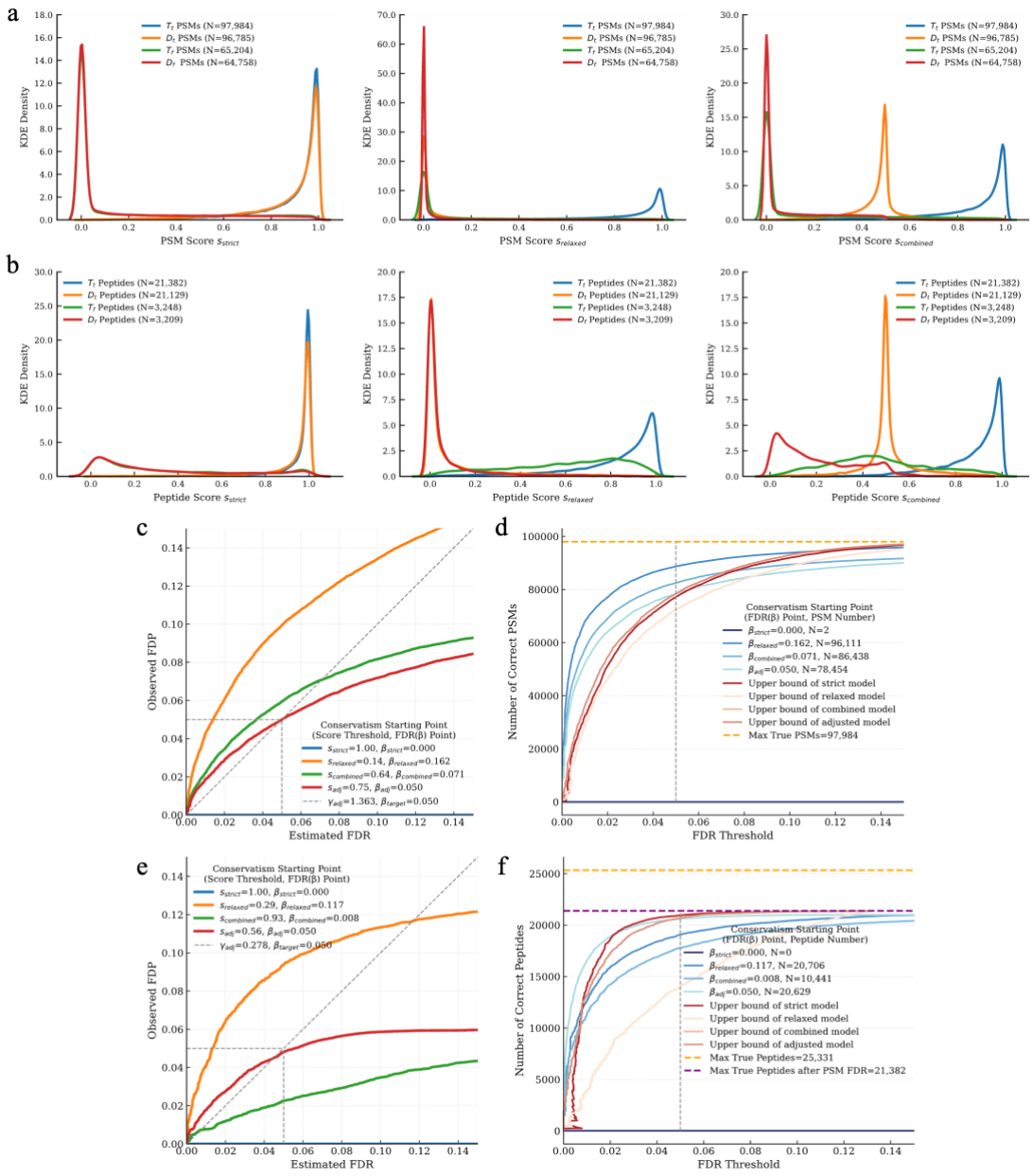


Figure S22: Dual-level CCT QC on *M. mazei* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

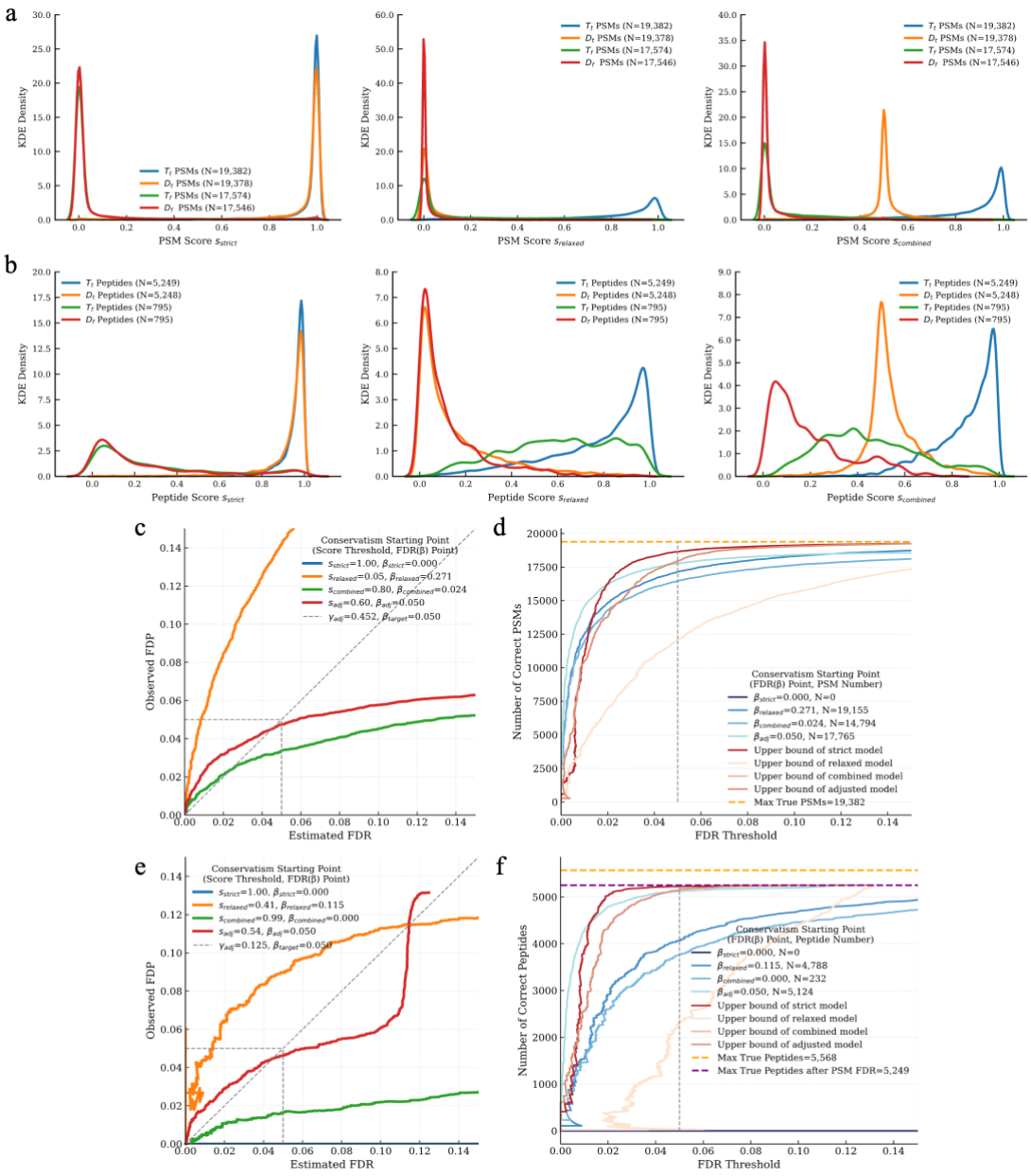


Figure S23: Dual-level CCT QC on *M. musculus* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

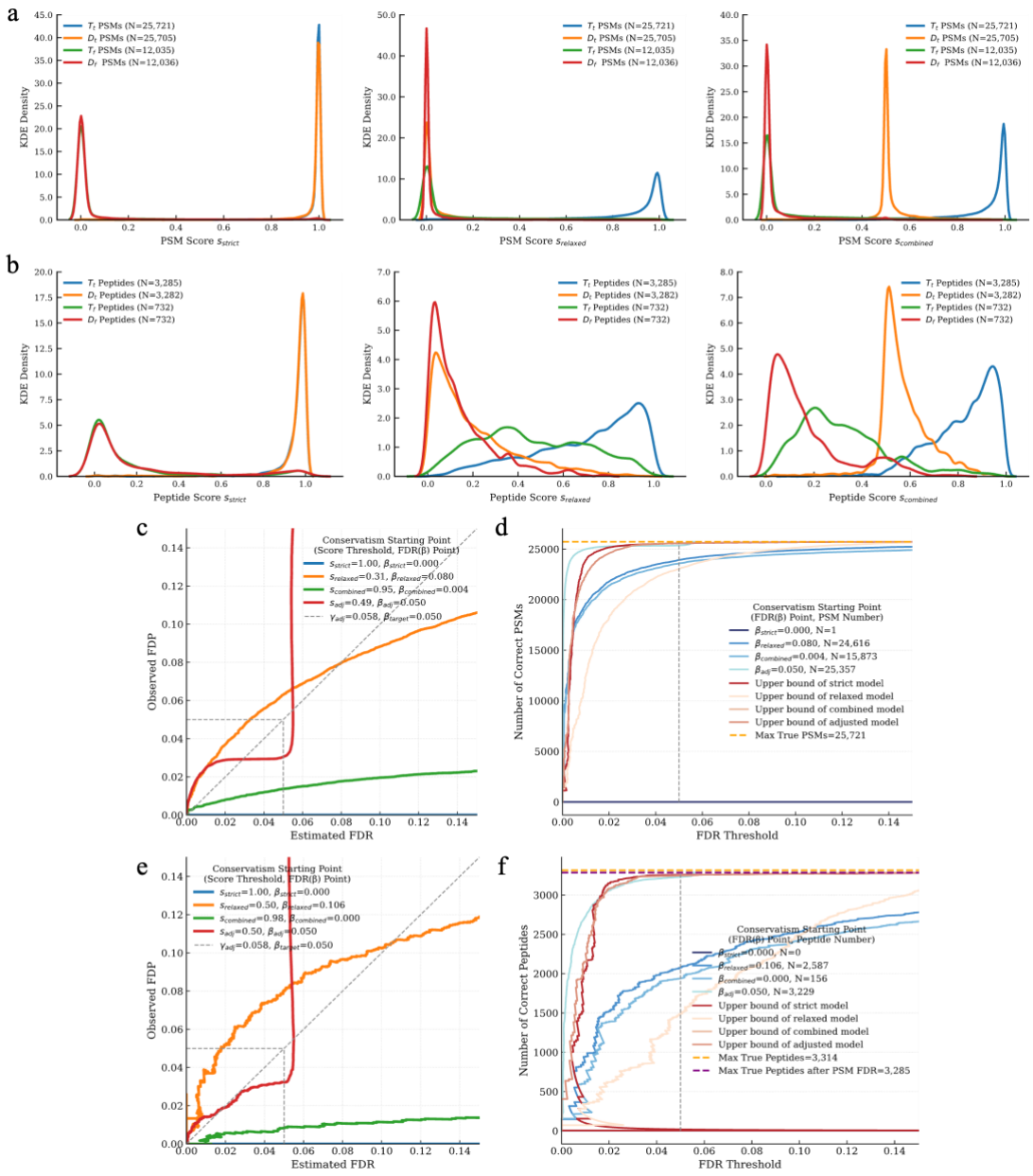


Figure S24: Dual-level CCT QC on *V. mungo* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

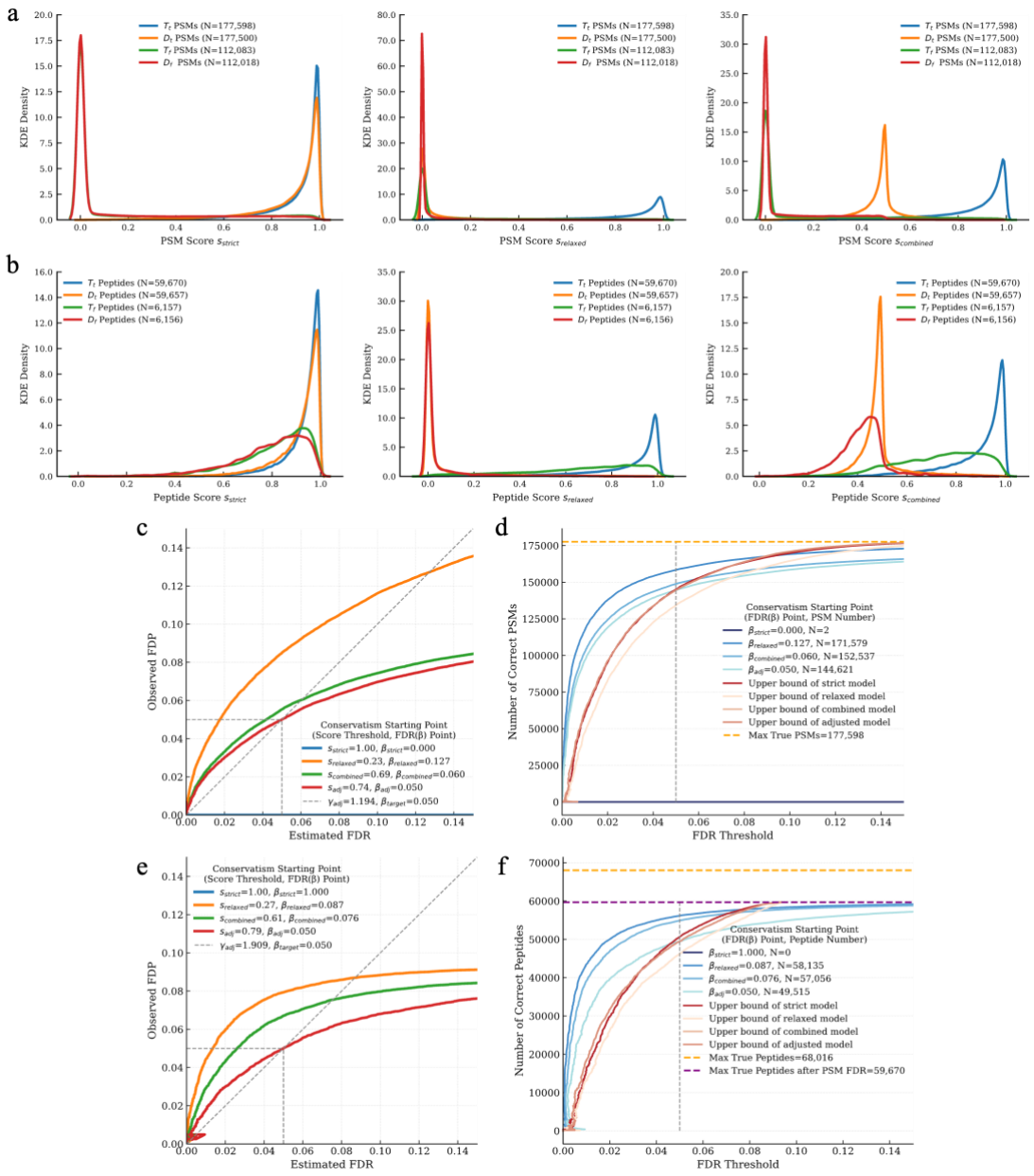


Figure S25: Dual-level CCT QC on *S. lycopersicum* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

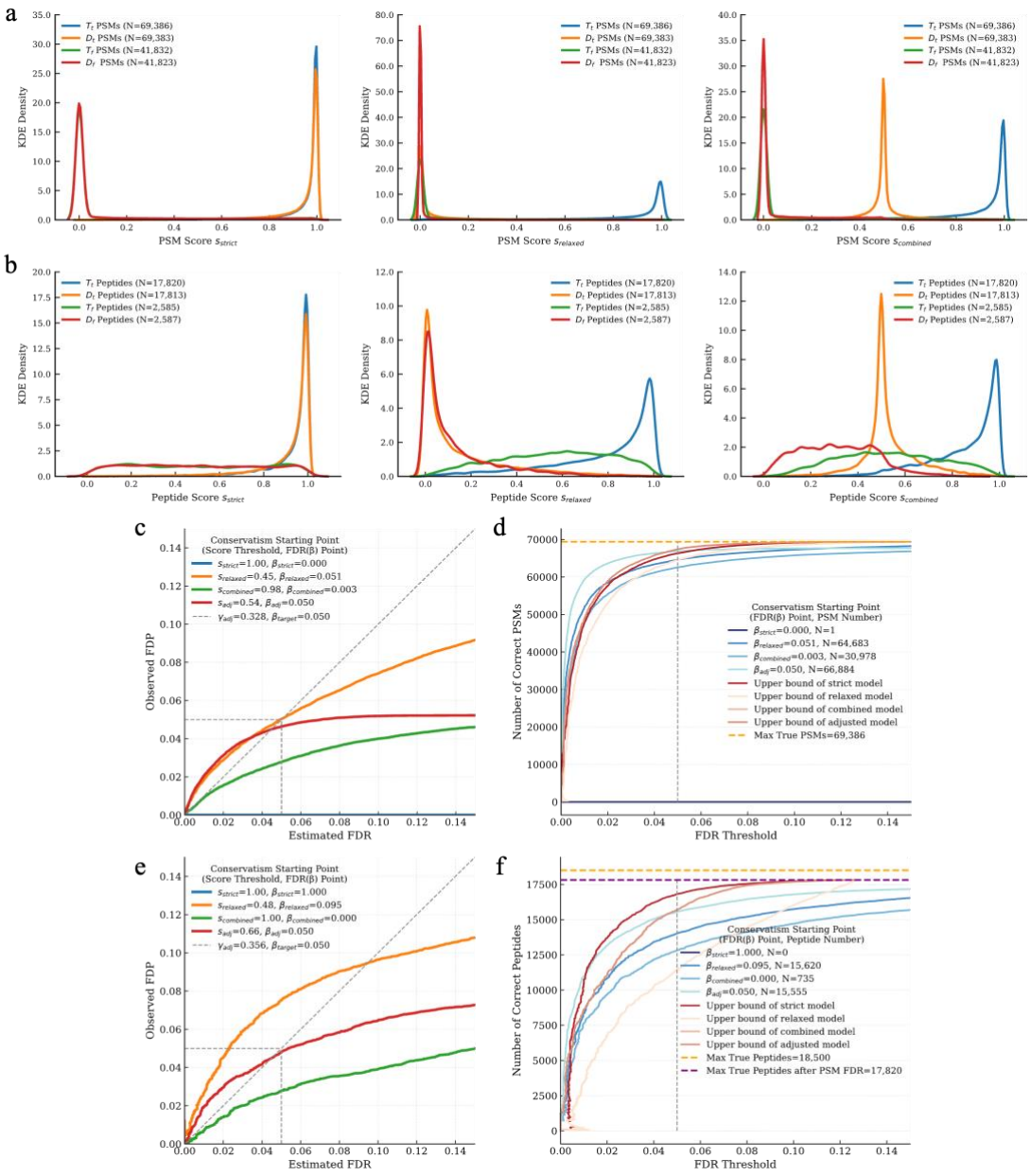


Figure S26: Dual-level CCT QC on *S. cerevisiae* test set using π-xNovo-DDA trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

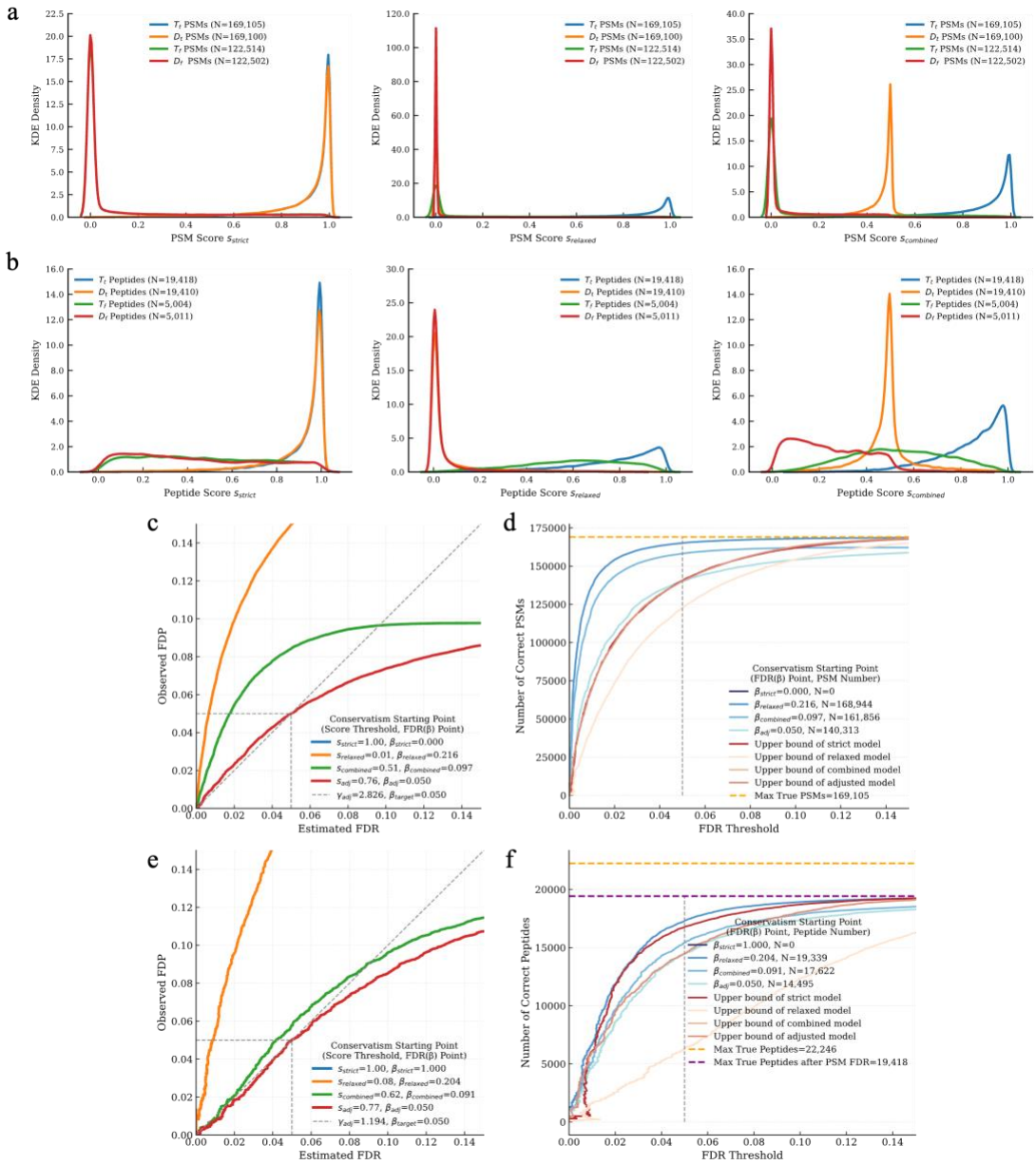


Figure S27: Dual-level CCT QC on *Bacillus* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

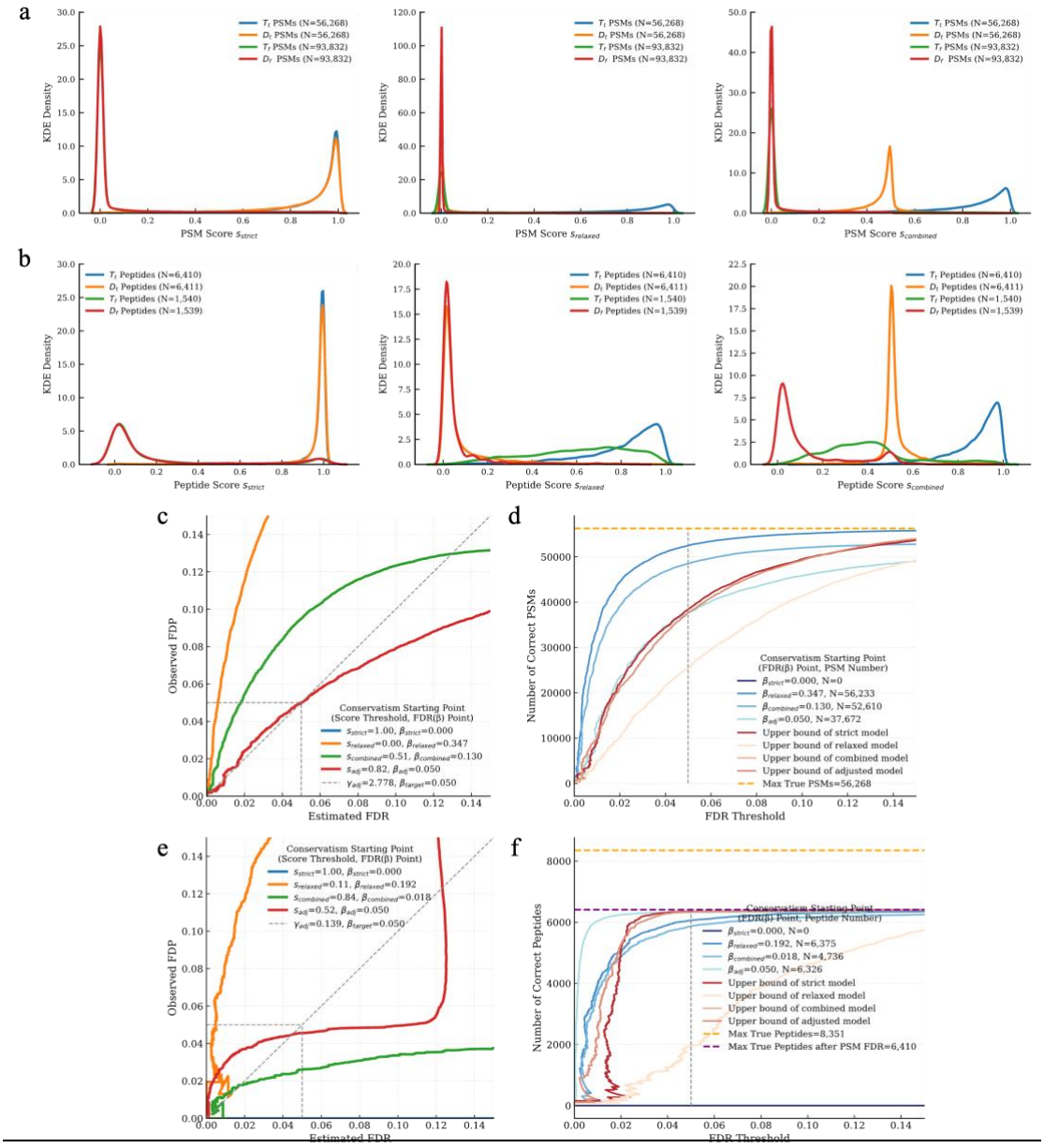


Figure S28: Dual-level CCT QC on *C. endoloripes* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

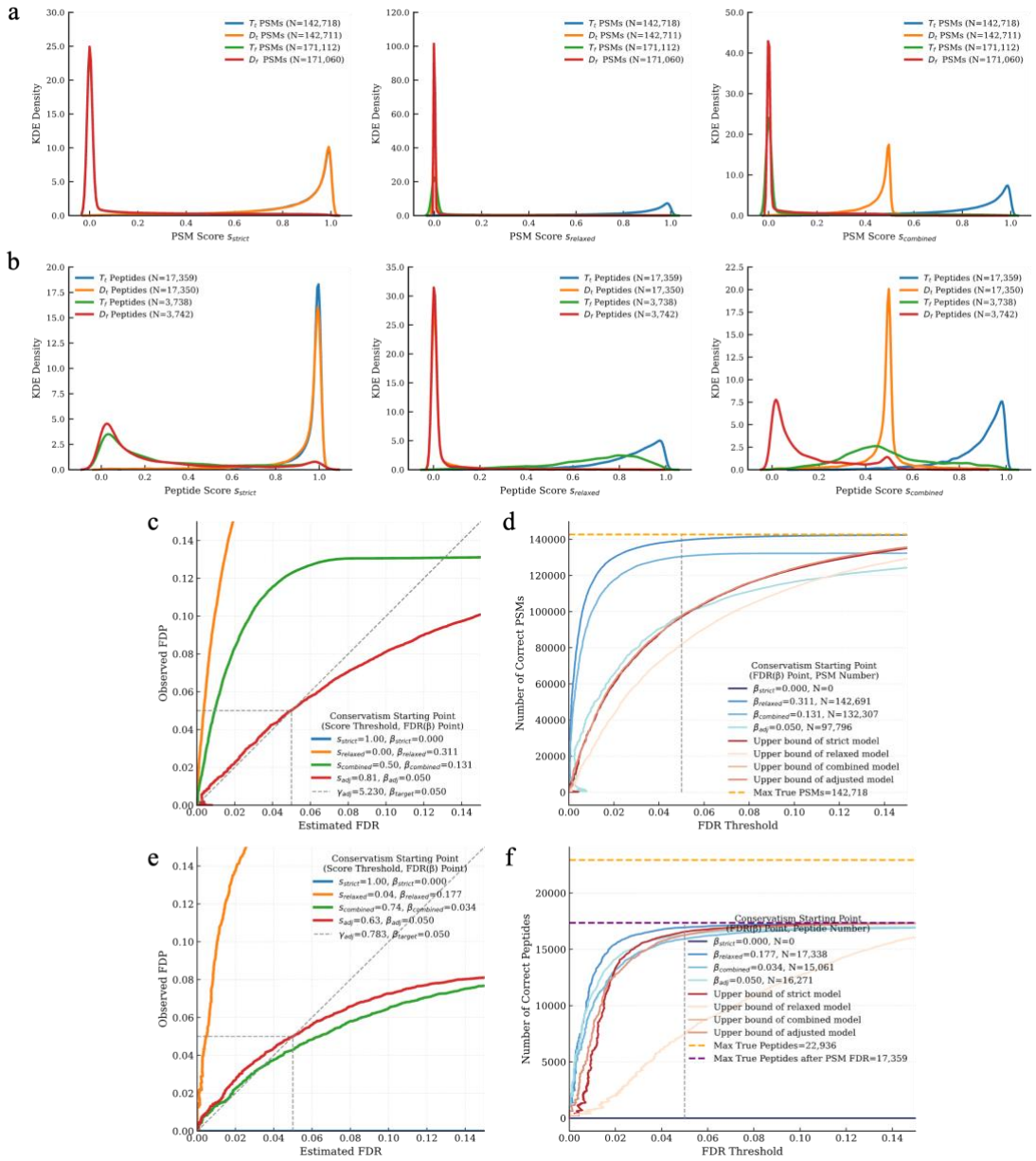


Figure S29: Dual-level CCT QC on *A. mellifera* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

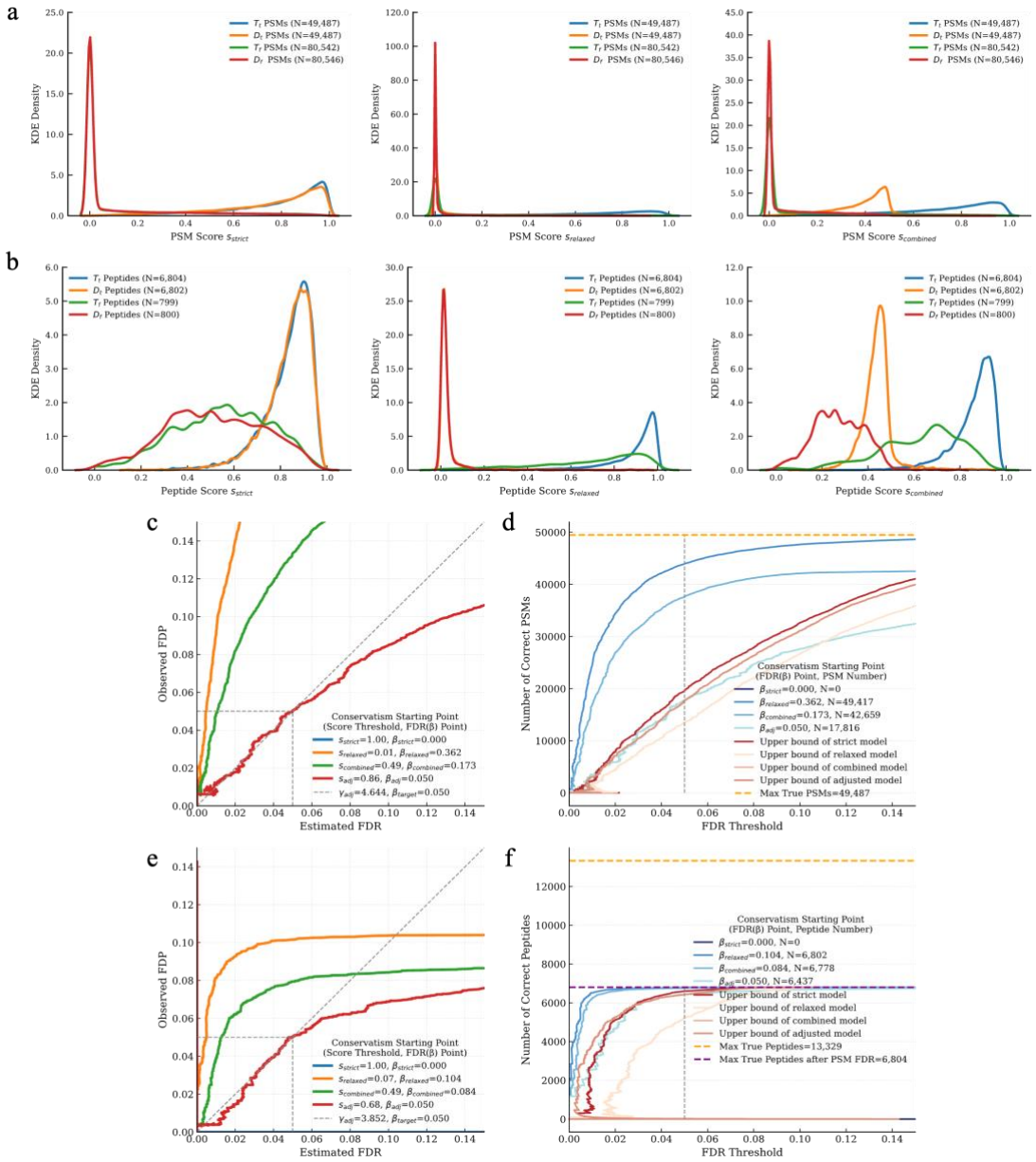


Figure S30: Dual-level CCT QC on *H. sapiens* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

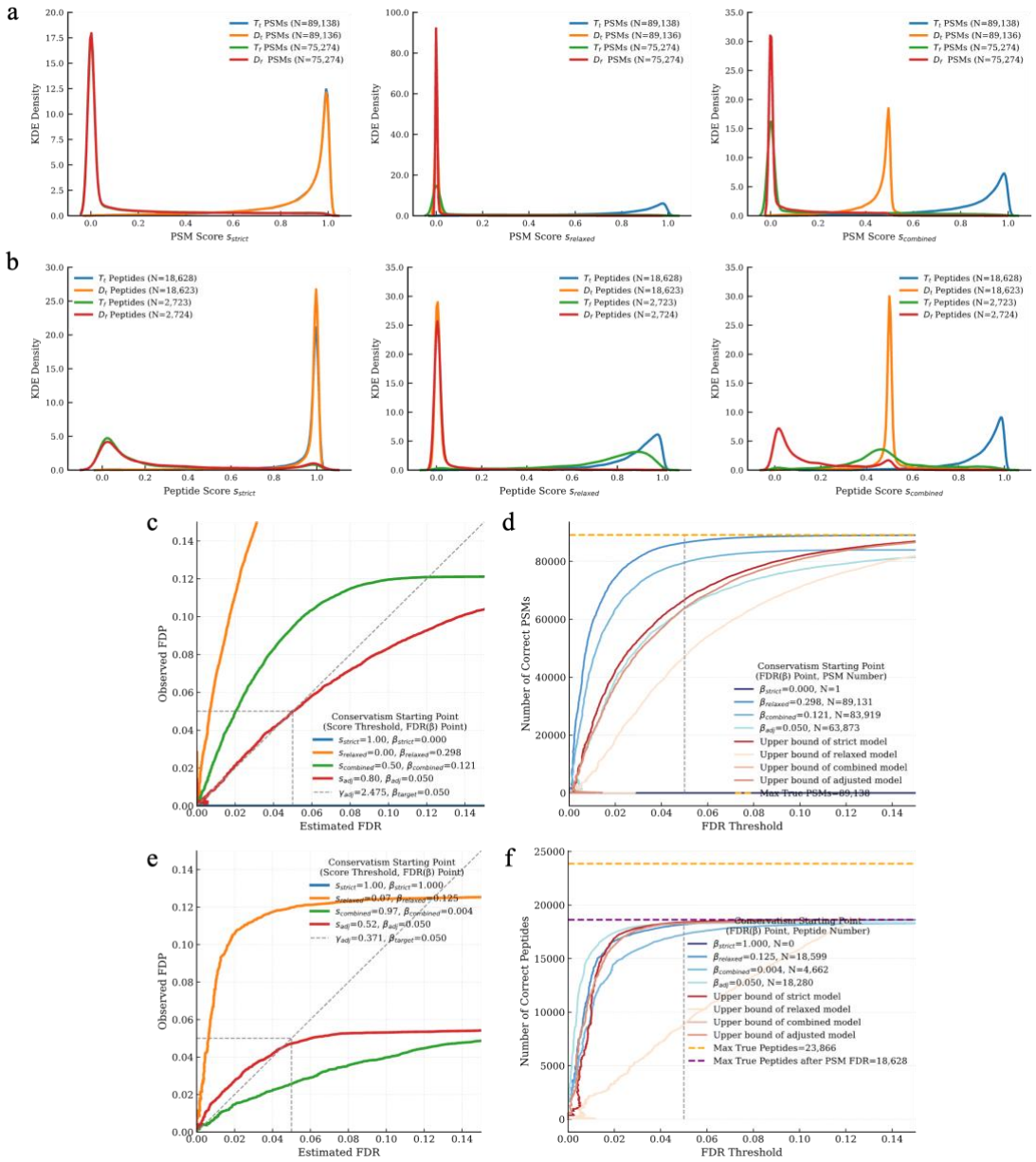


Figure S31: Dual-level CCT QC on *M. mazei* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

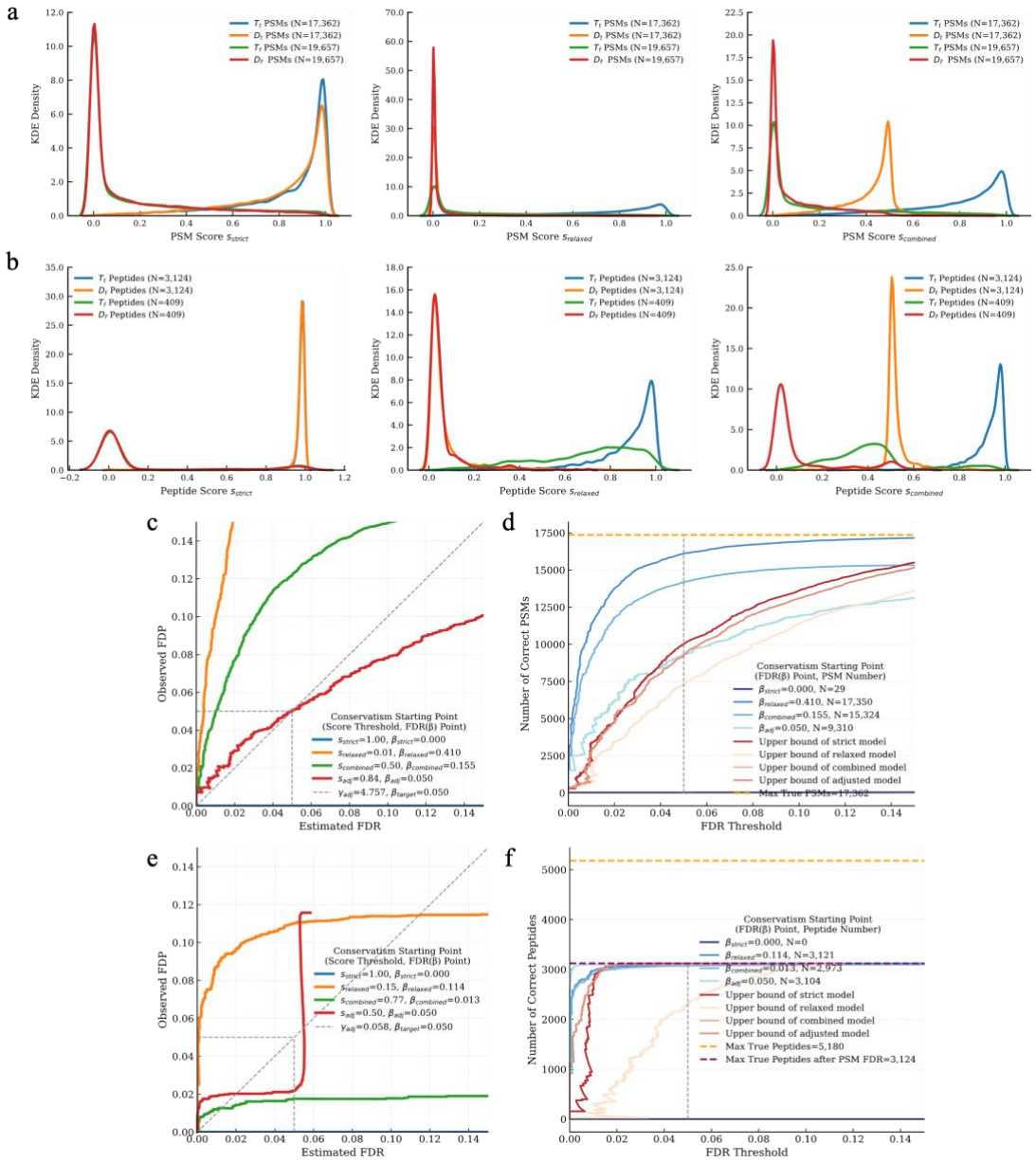


Figure S32: Dual-level CCT QC on *M. musculus* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

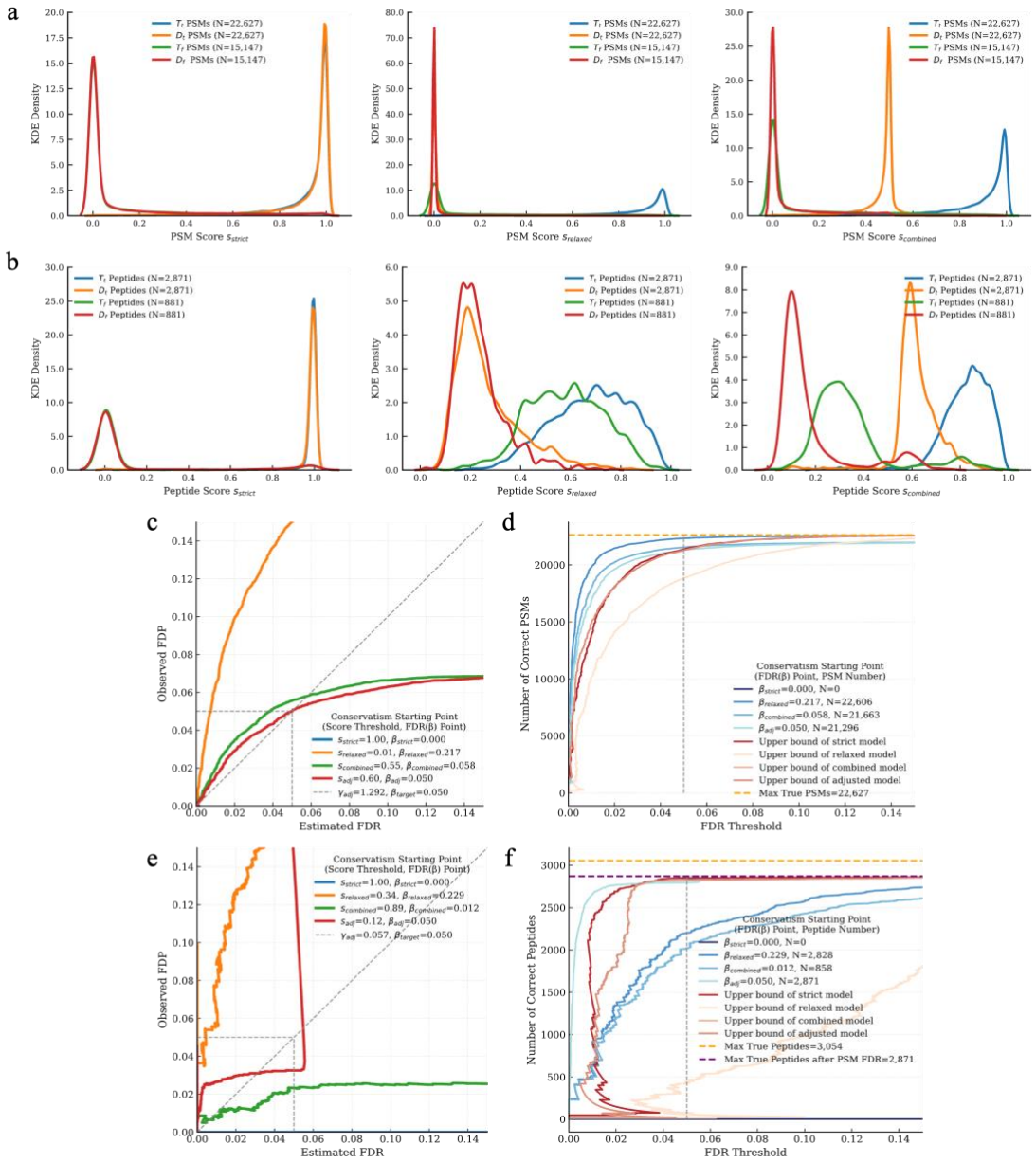


Figure S33: Dual-level CCT QC on *V. mungo* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

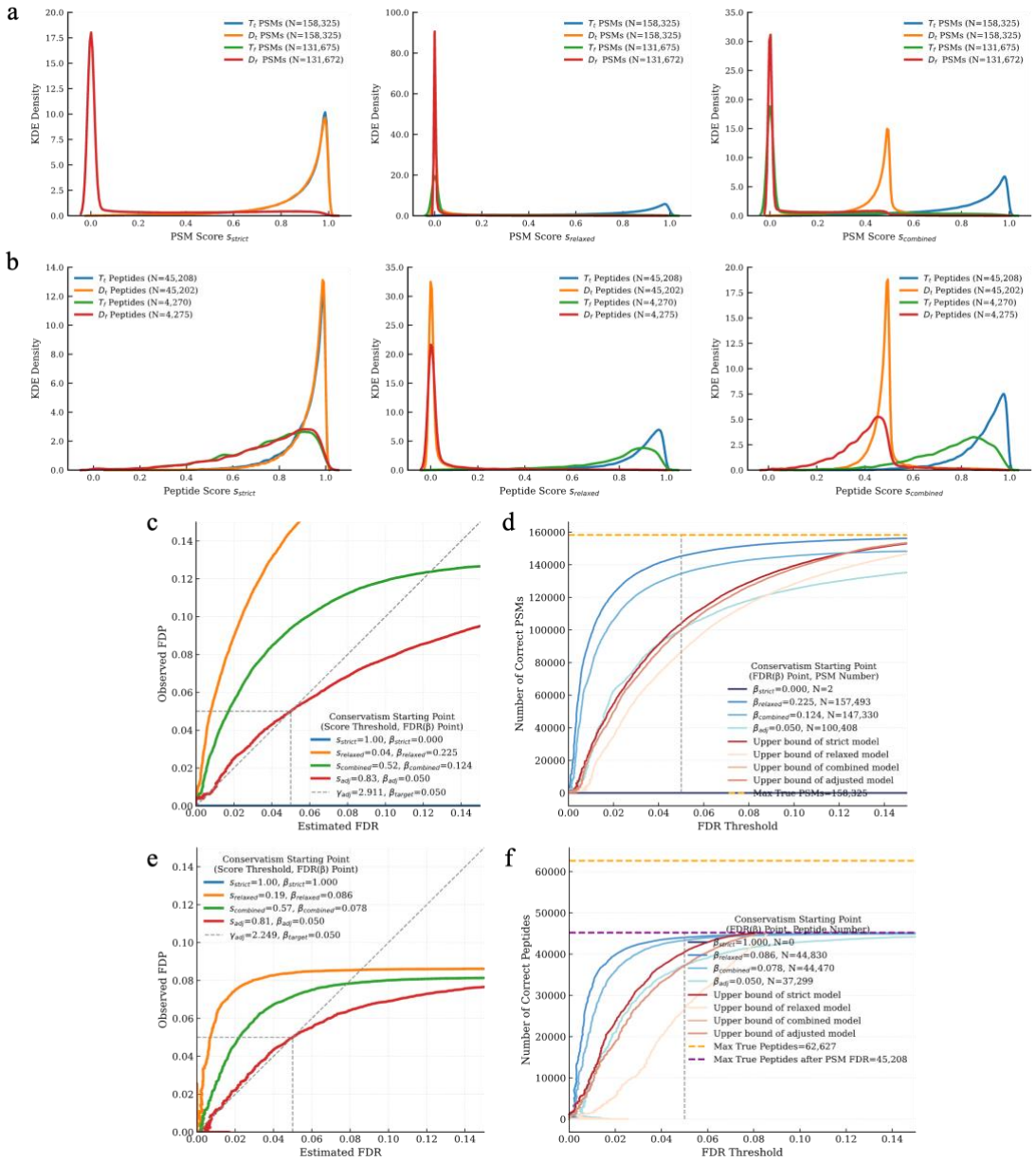


Figure S34: Dual-level CCT QC on *S. lycopersicum* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

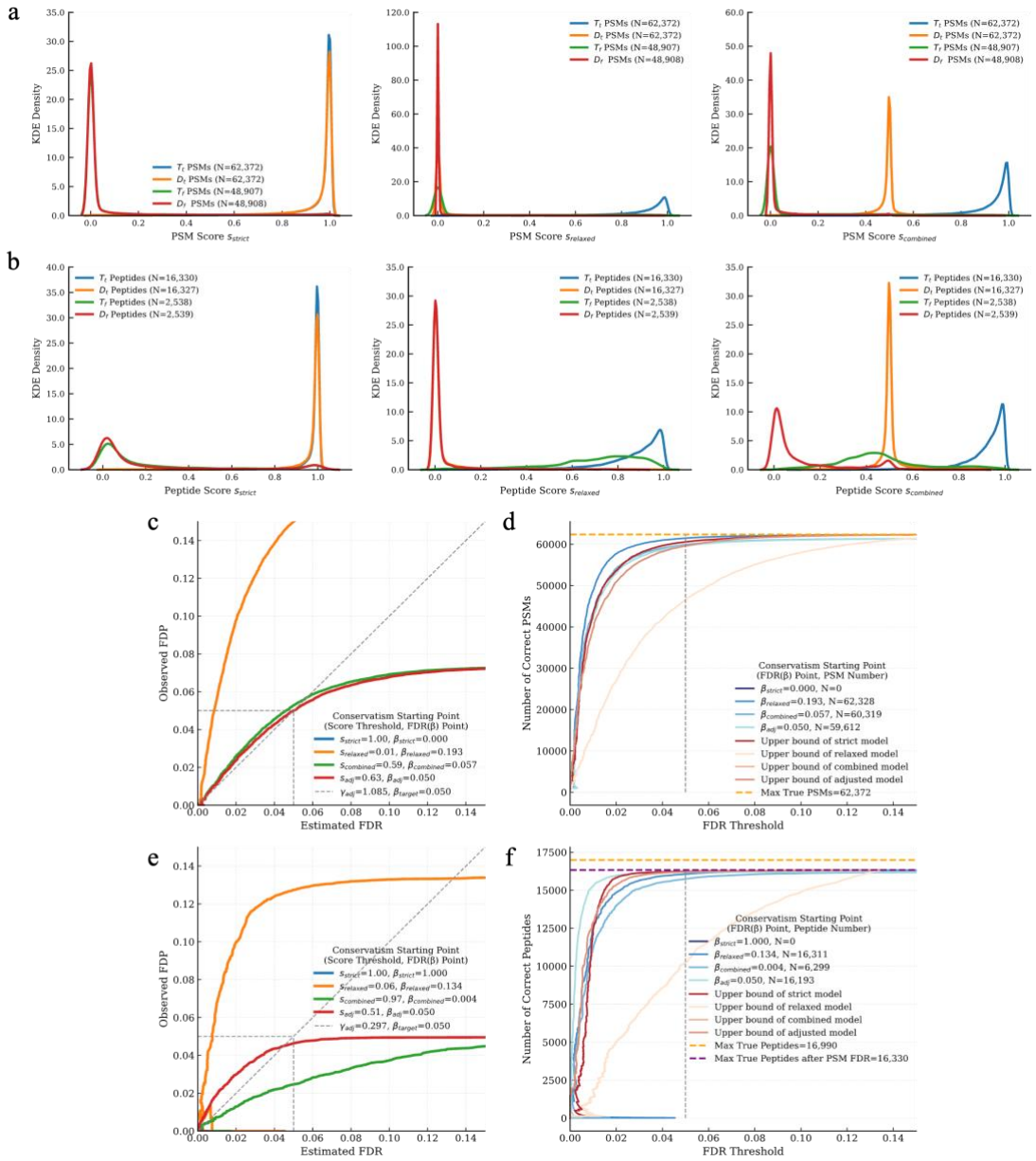


Figure S35: Dual-level CCT QC on *S. cerevisiae* test set using π-HelixNovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

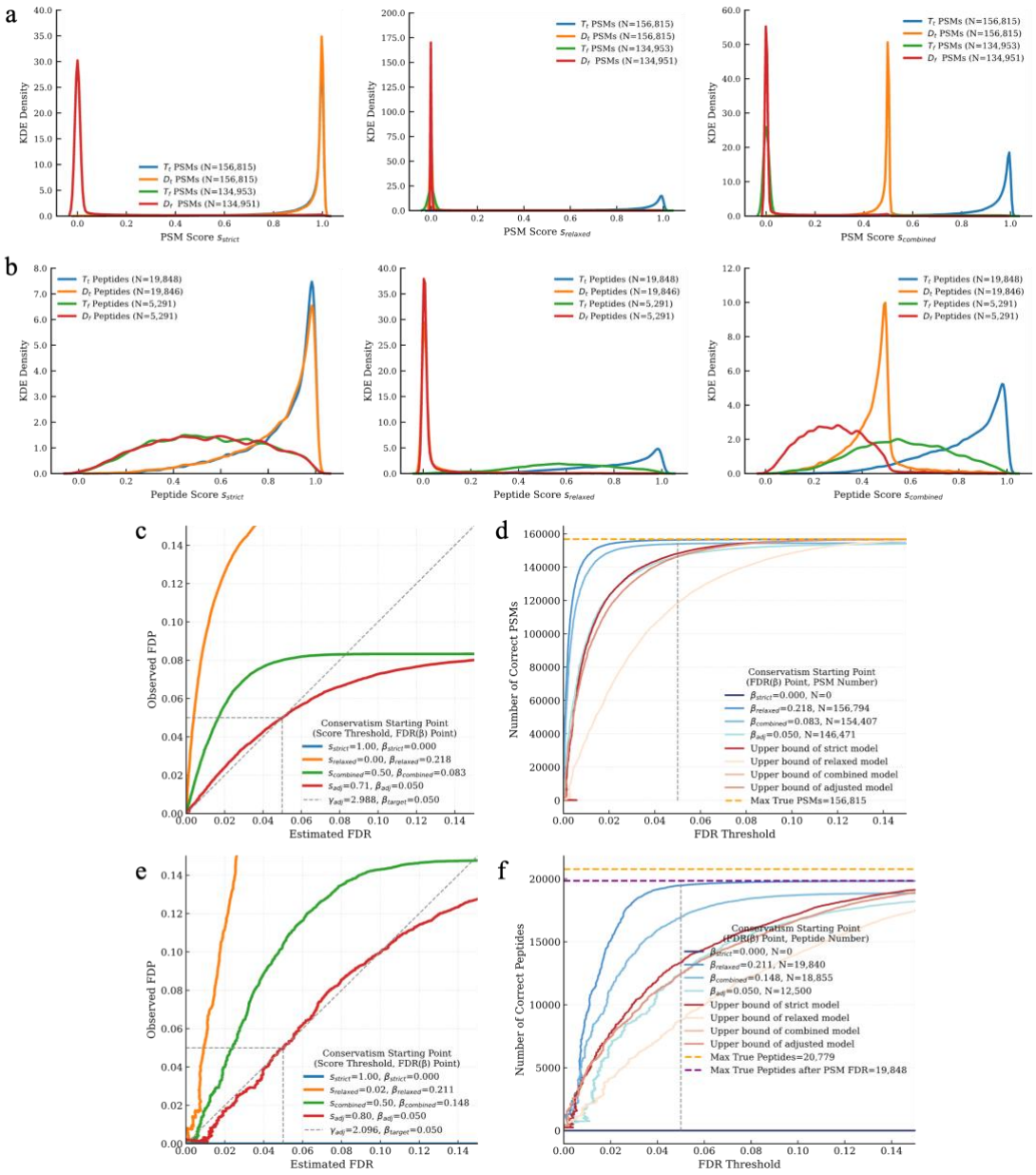


Figure S36: Dual-level CCT QC on *Bacillus* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

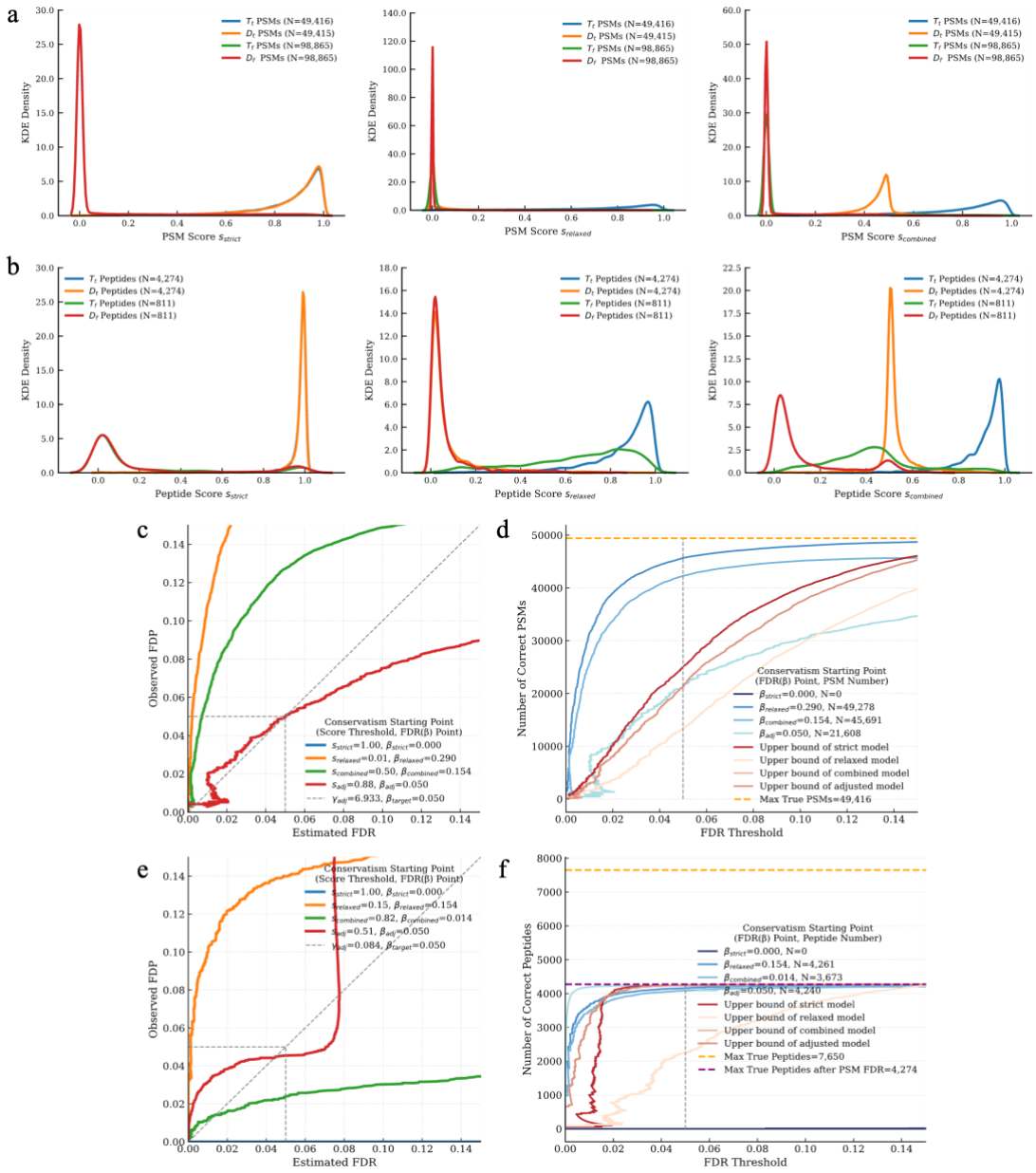


Figure S37: Dual-level CCT QC on *C. endoloripes* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

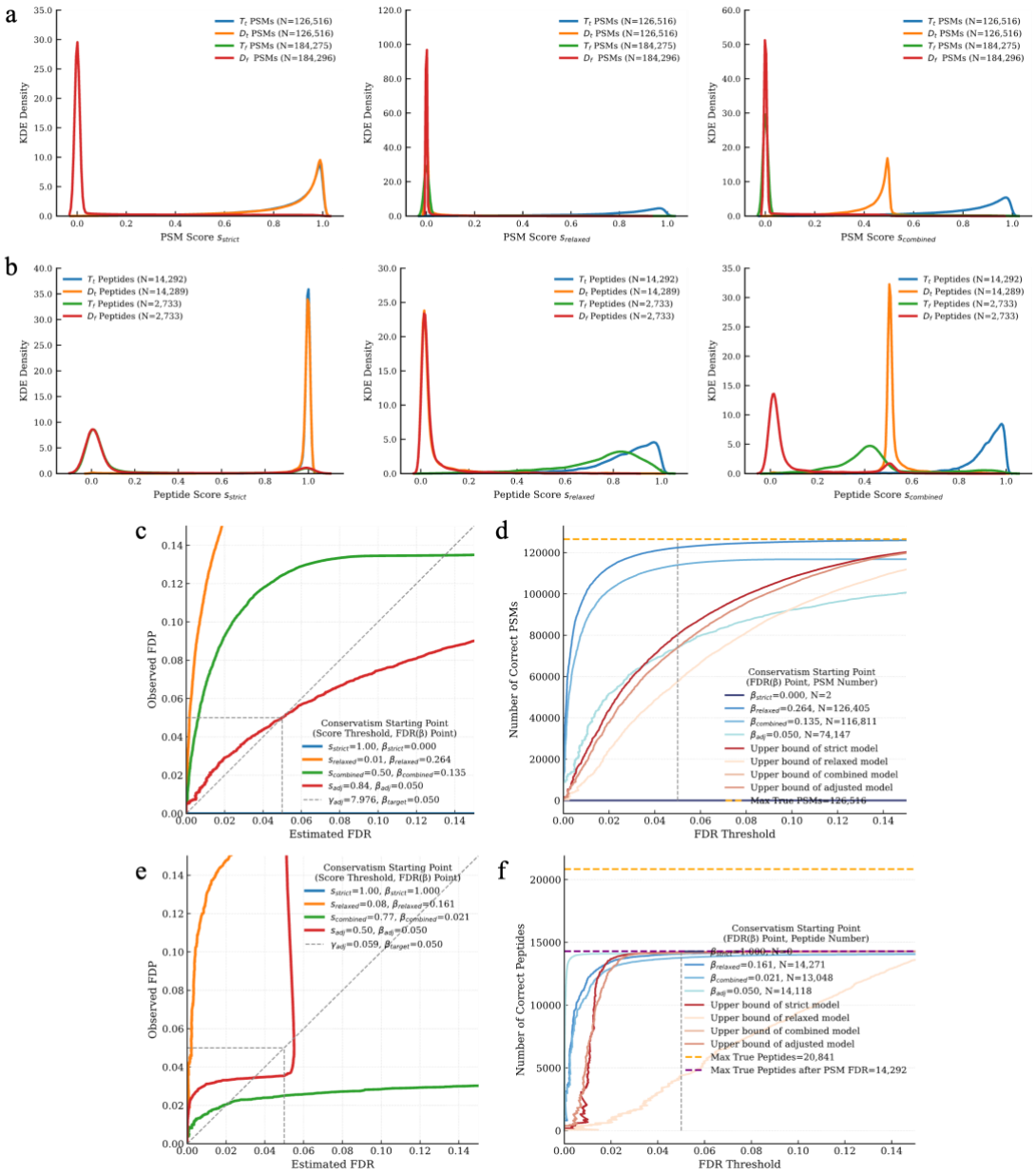


Figure S38: Dual-level CCT QC on *A. mellifera* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

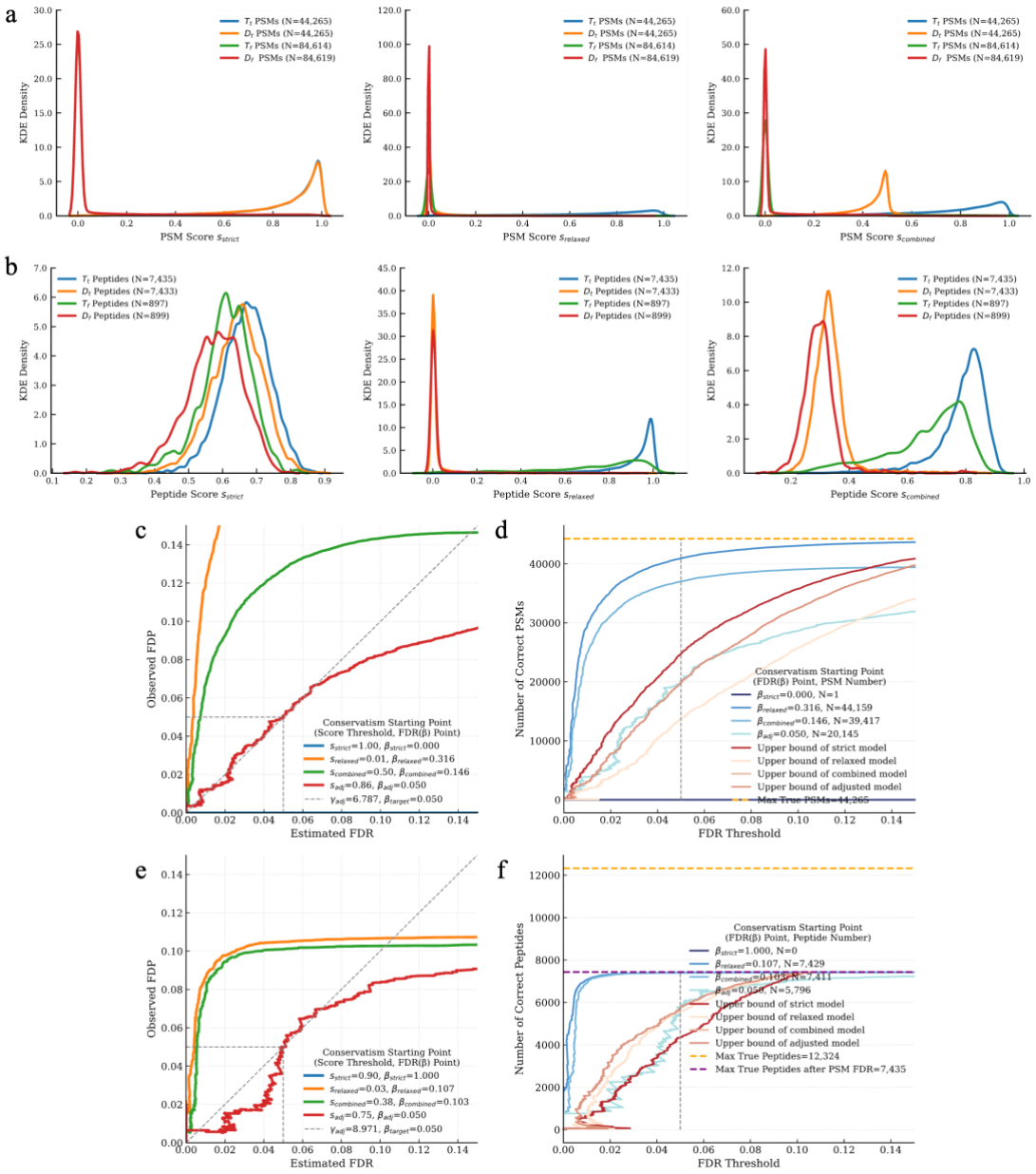


Figure S39: Dual-level CCT QC on *H. sapiens* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

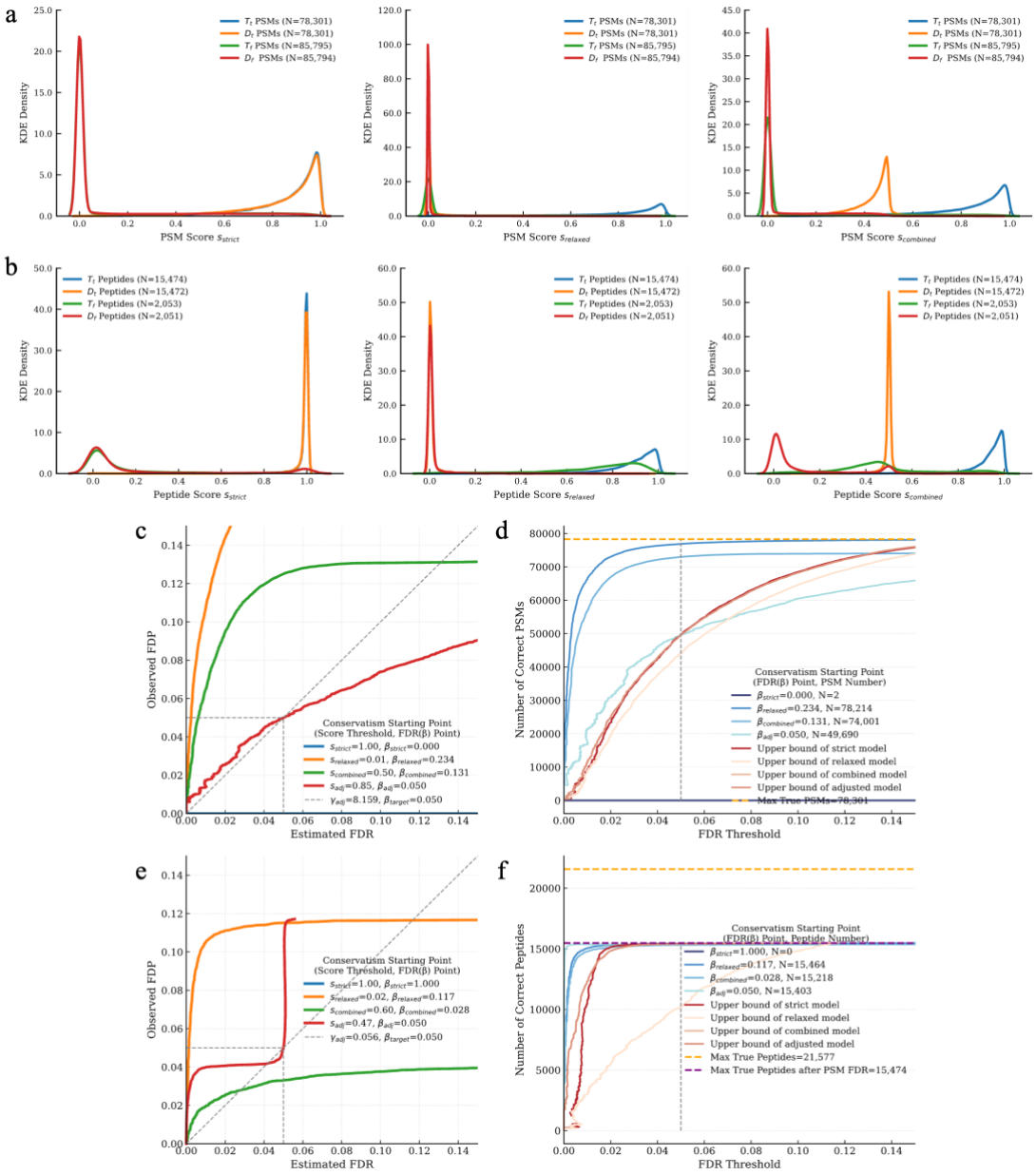


Figure S40: Dual-level CCT QC on *M. mazei* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

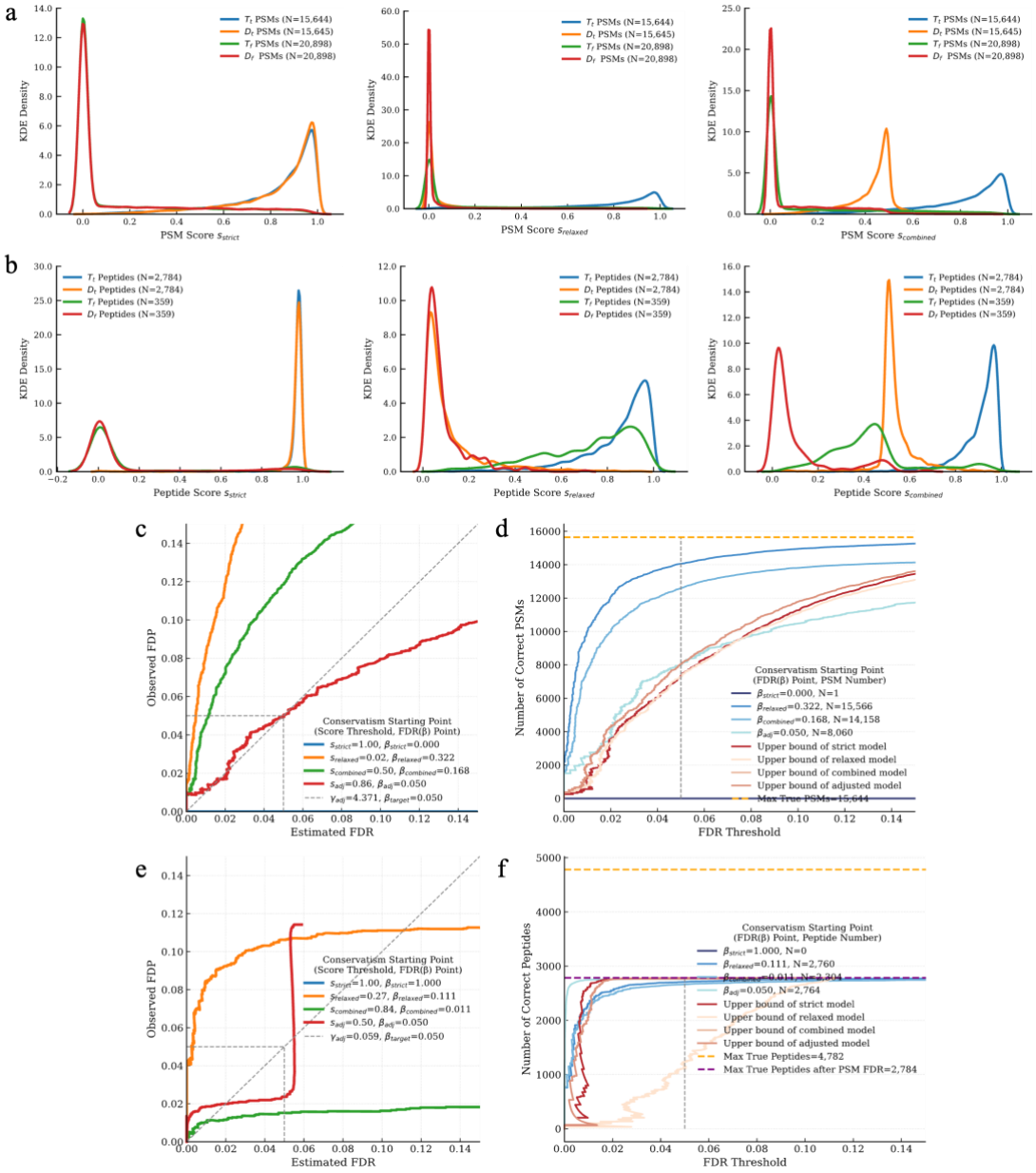


Figure S41: Dual-level CCT QC on *M. musculus* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

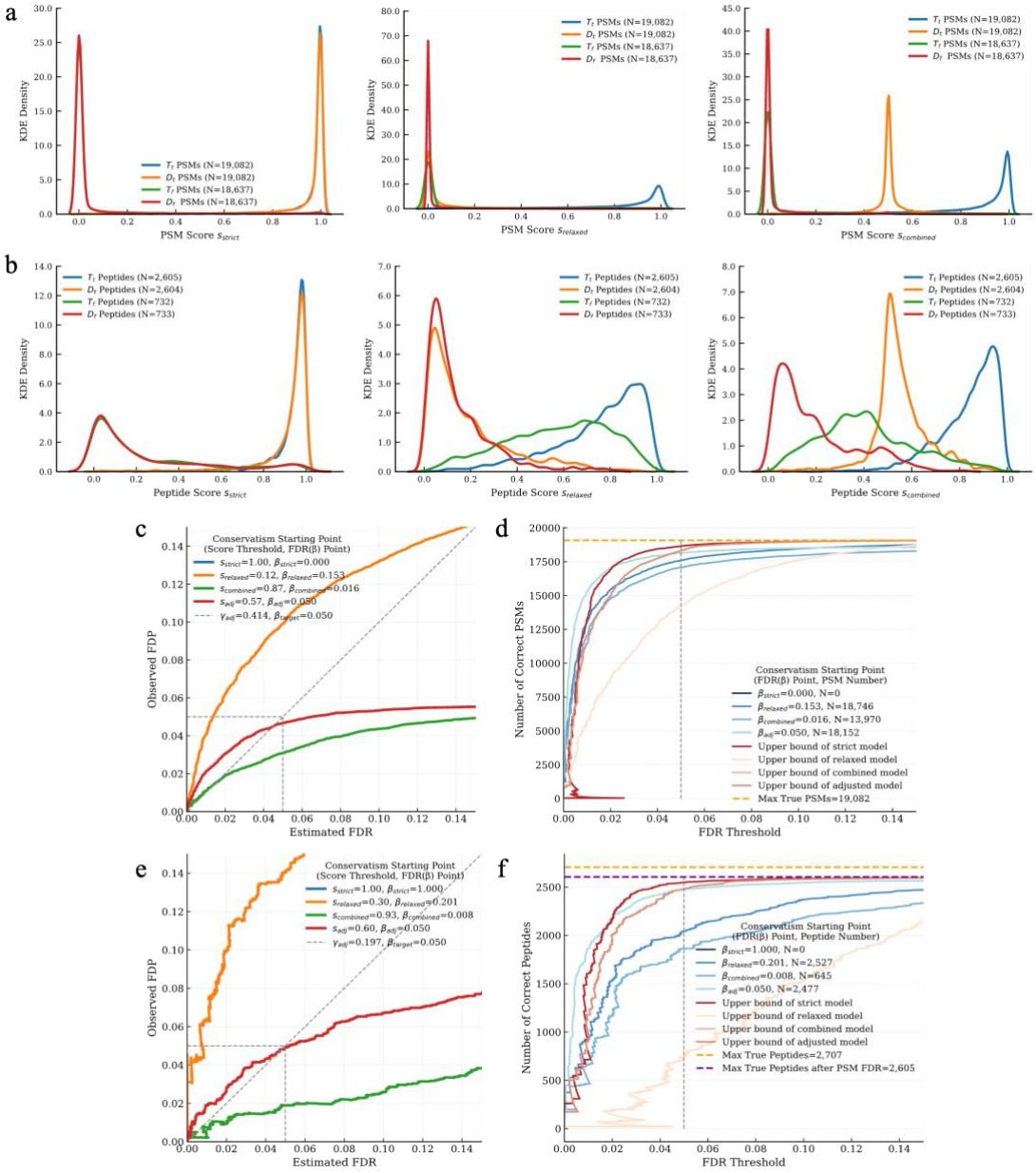


Figure S42: Dual-level CCT QC on *V. mungo* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

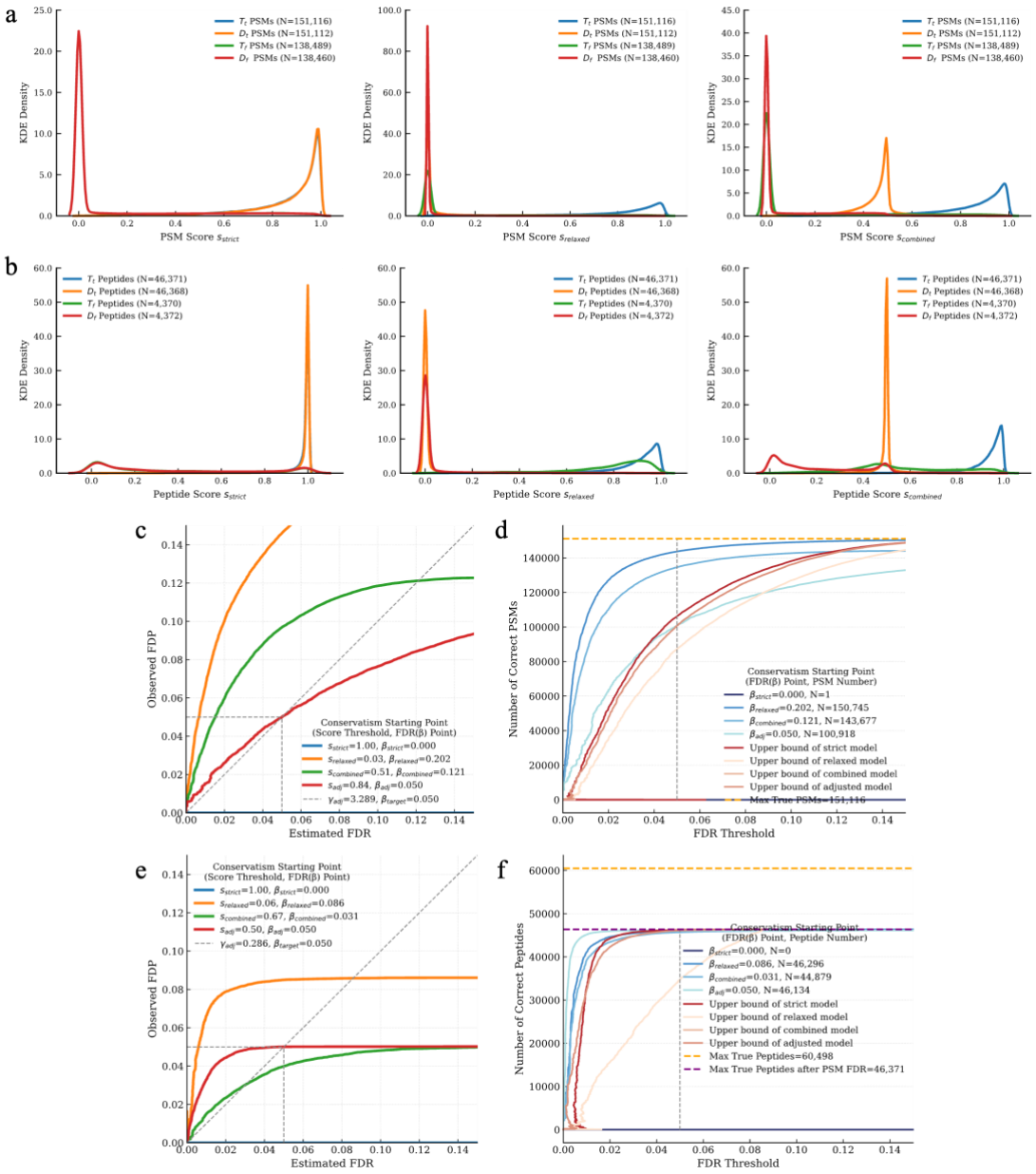


Figure S43: Dual-level CCT QC on *S. lycopersicum* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

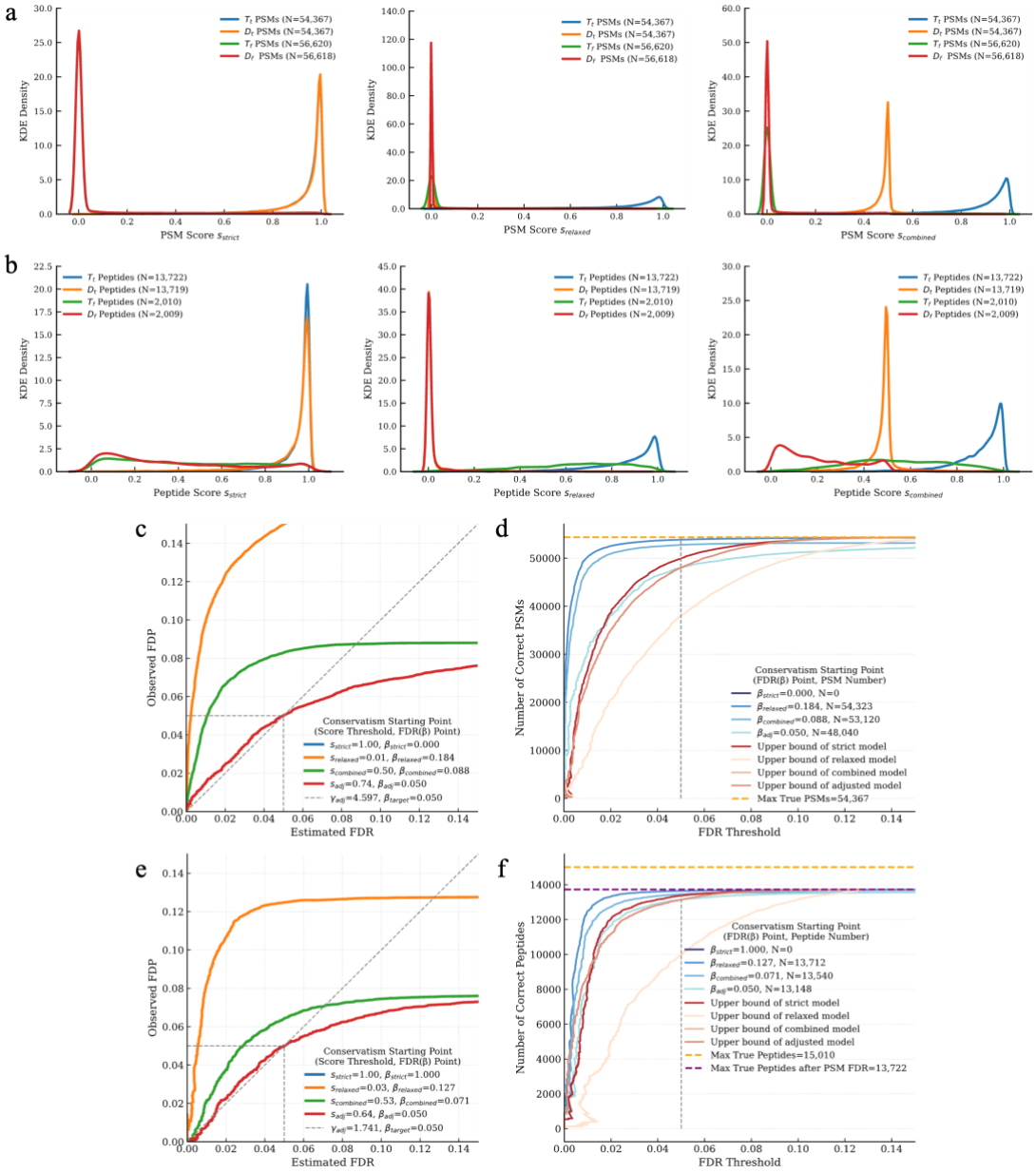


Figure S44: Dual-level CCT QC on *S. cerevisiae* test set using Casanovo trained on the nine-species training set. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

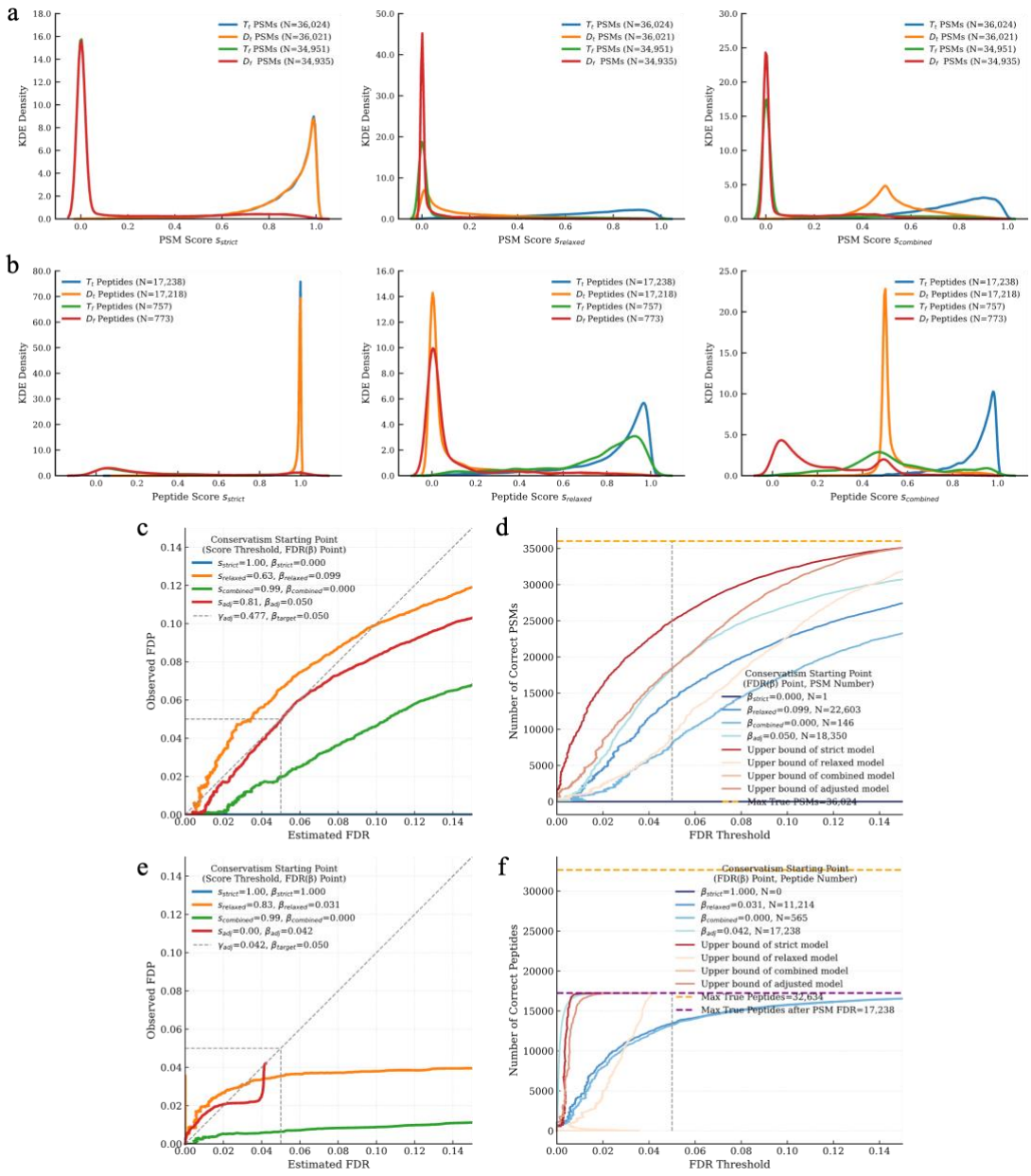


Figure S45: Dual-level CCT QC on replicate 1 of Astral DDA dataset using π-xNovo-DDA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

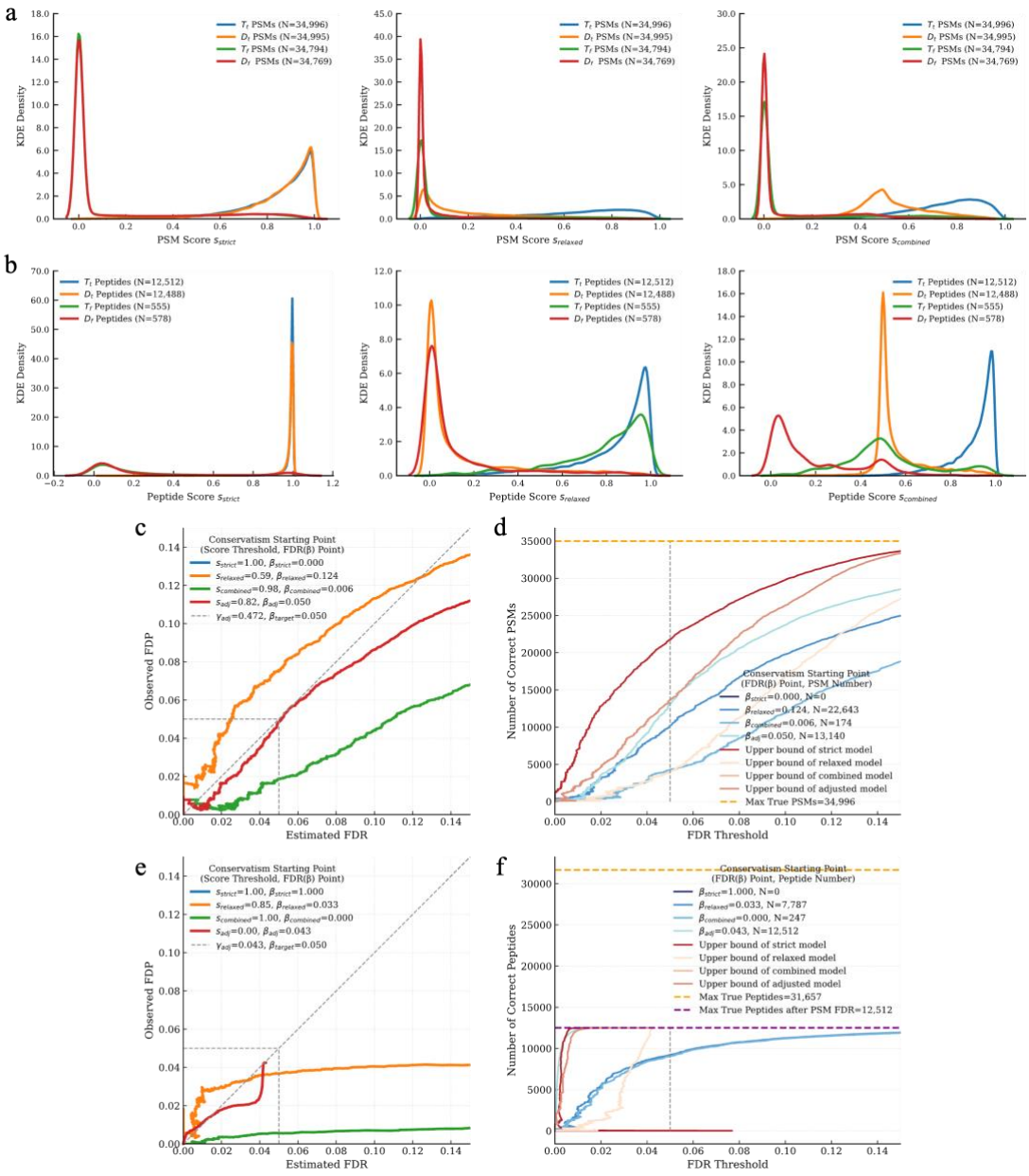


Figure S46: Dual-level CCT QC on replicate 2 of Astral DDA dataset using π-xNovo-DDA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

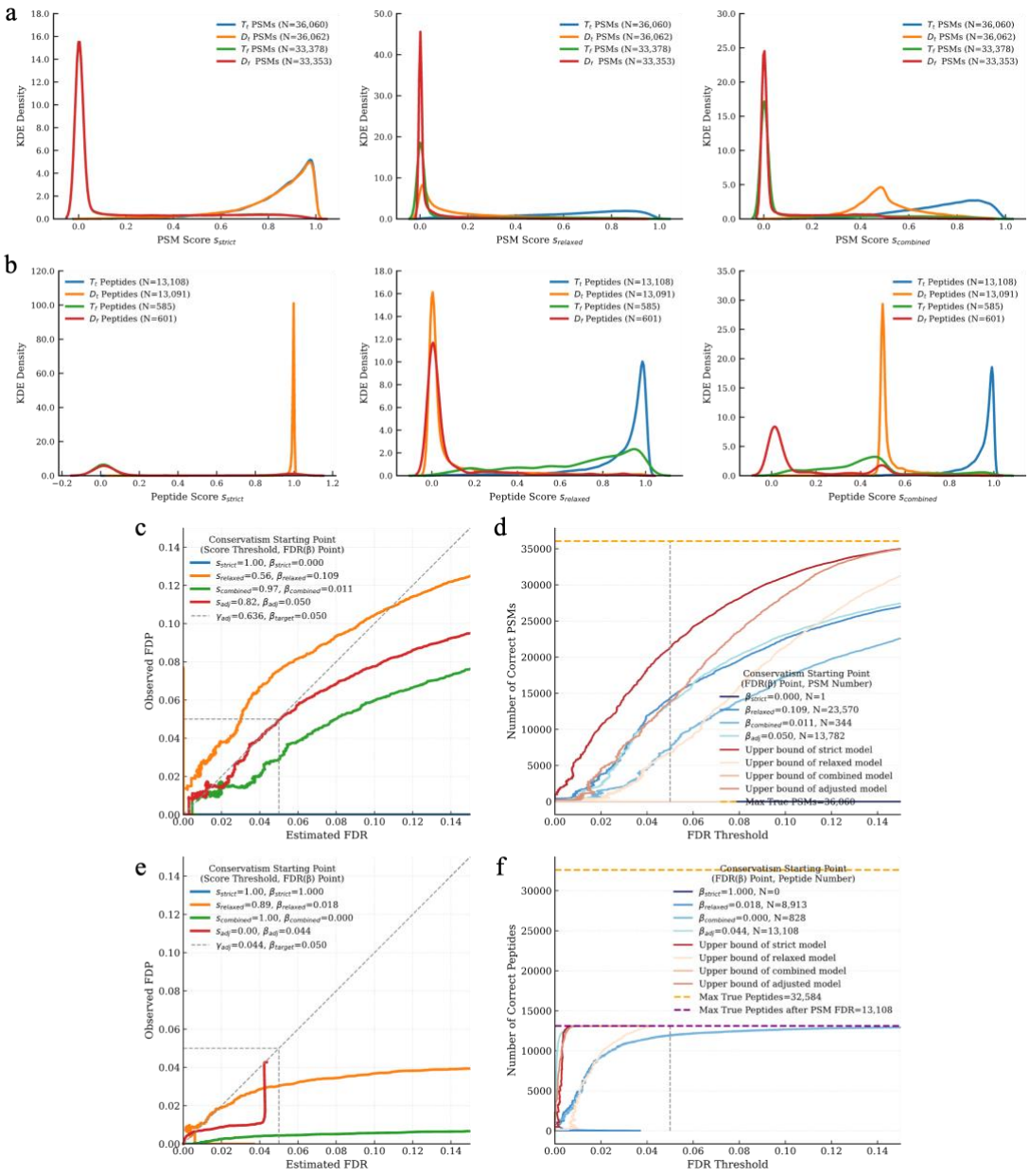


Figure S47: Dual-level CCT QC on replicate 3 of Astral DDA dataset using π-xNovo-DDA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

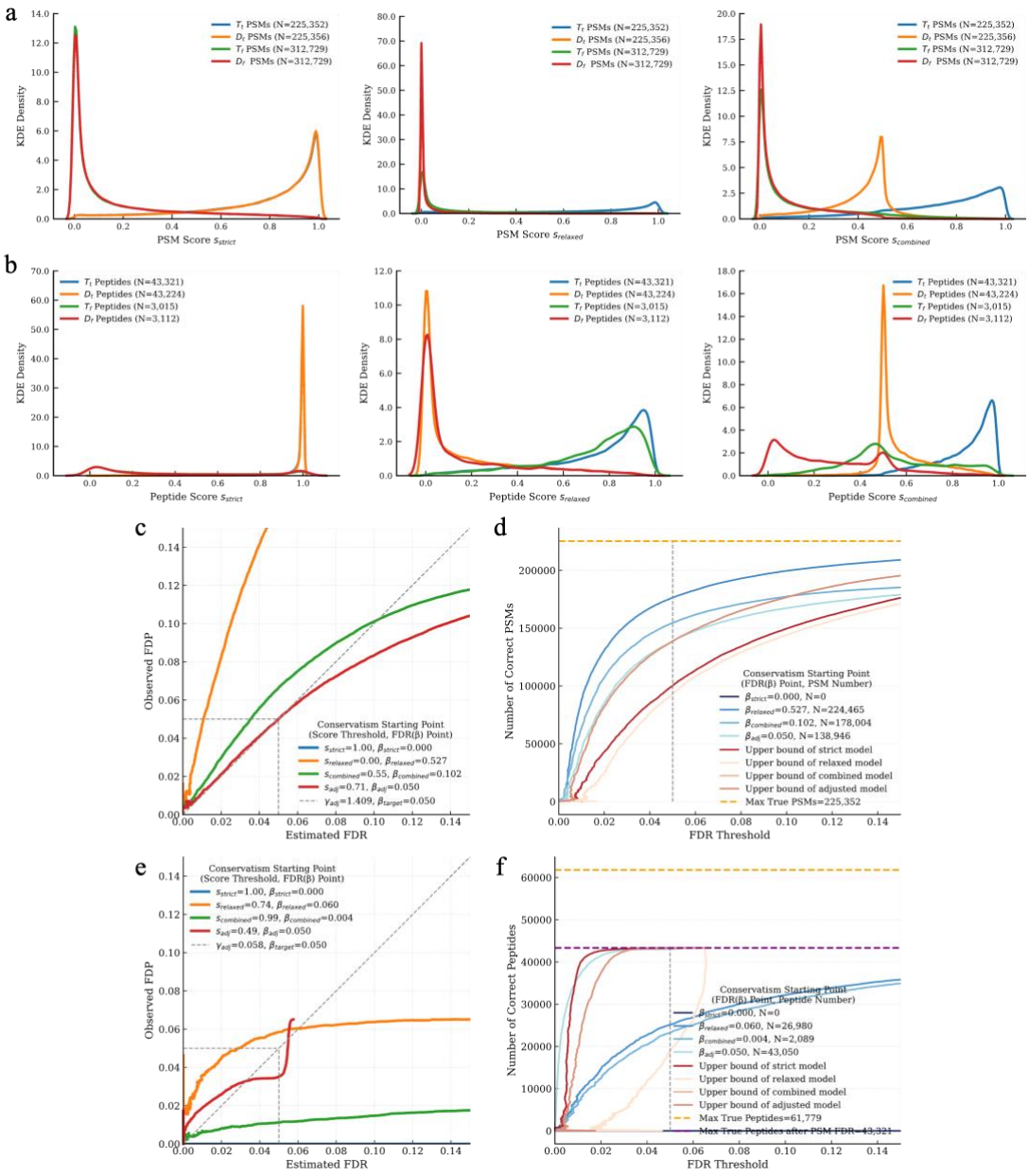


Figure S48: Dual-level CCT QC on replicate 1 of Astral DIA dataset (2p5ms) using π-xNovo-DIA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

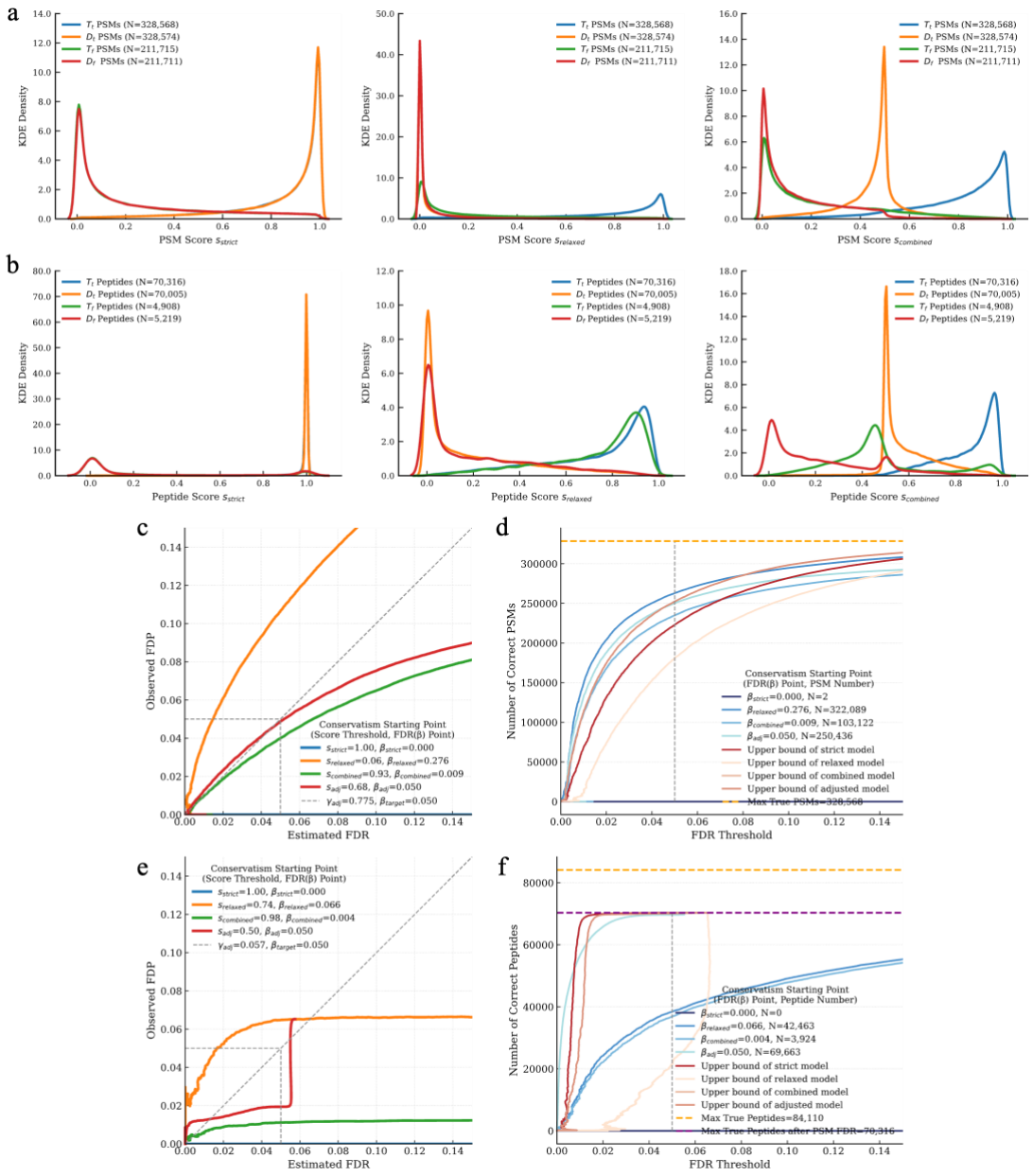


Figure S49: Dual-level CCT QC on replicate 2 of Astral DIA dataset (2p5ms) using π-xNovo-DIA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

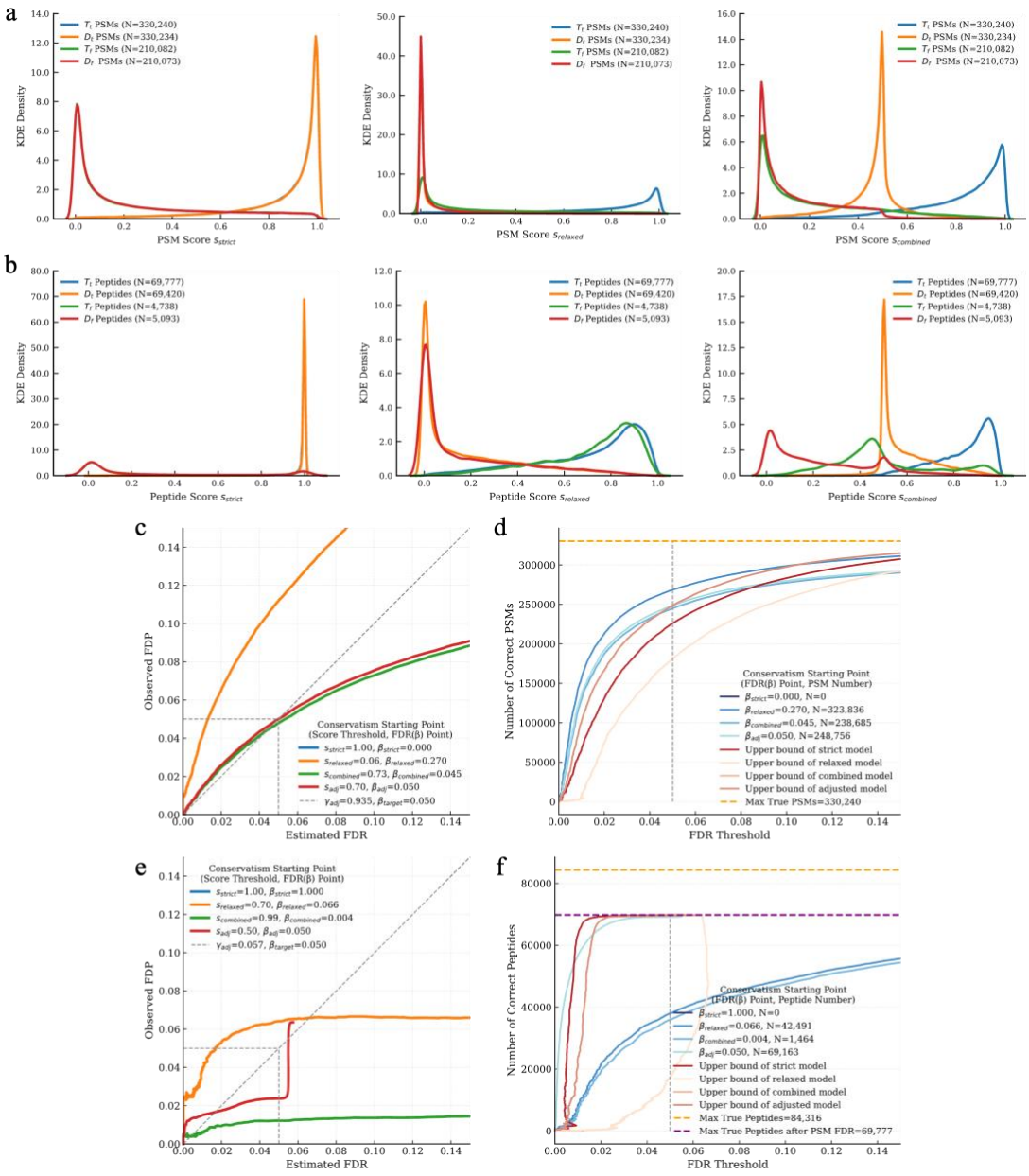


Figure S50: Dual-level CCT QC on replicate 3 of Astral DIA dataset (2p5ms) using π-xNovo-DIA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

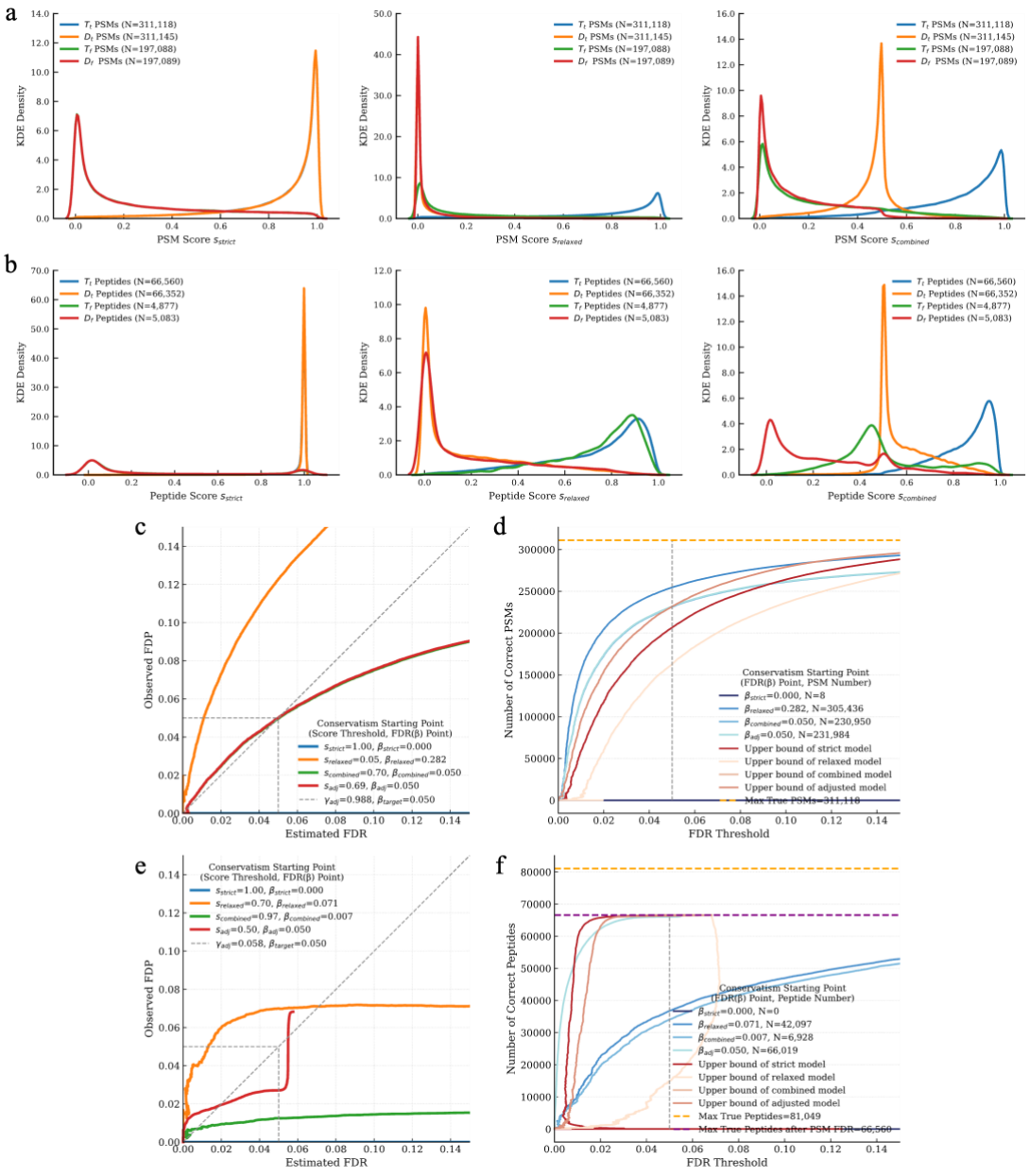


Figure S51: Dual-level CCT QC on replicate 1 of Astral DIA dataset (3p5ms) using π-xNovo-DIA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

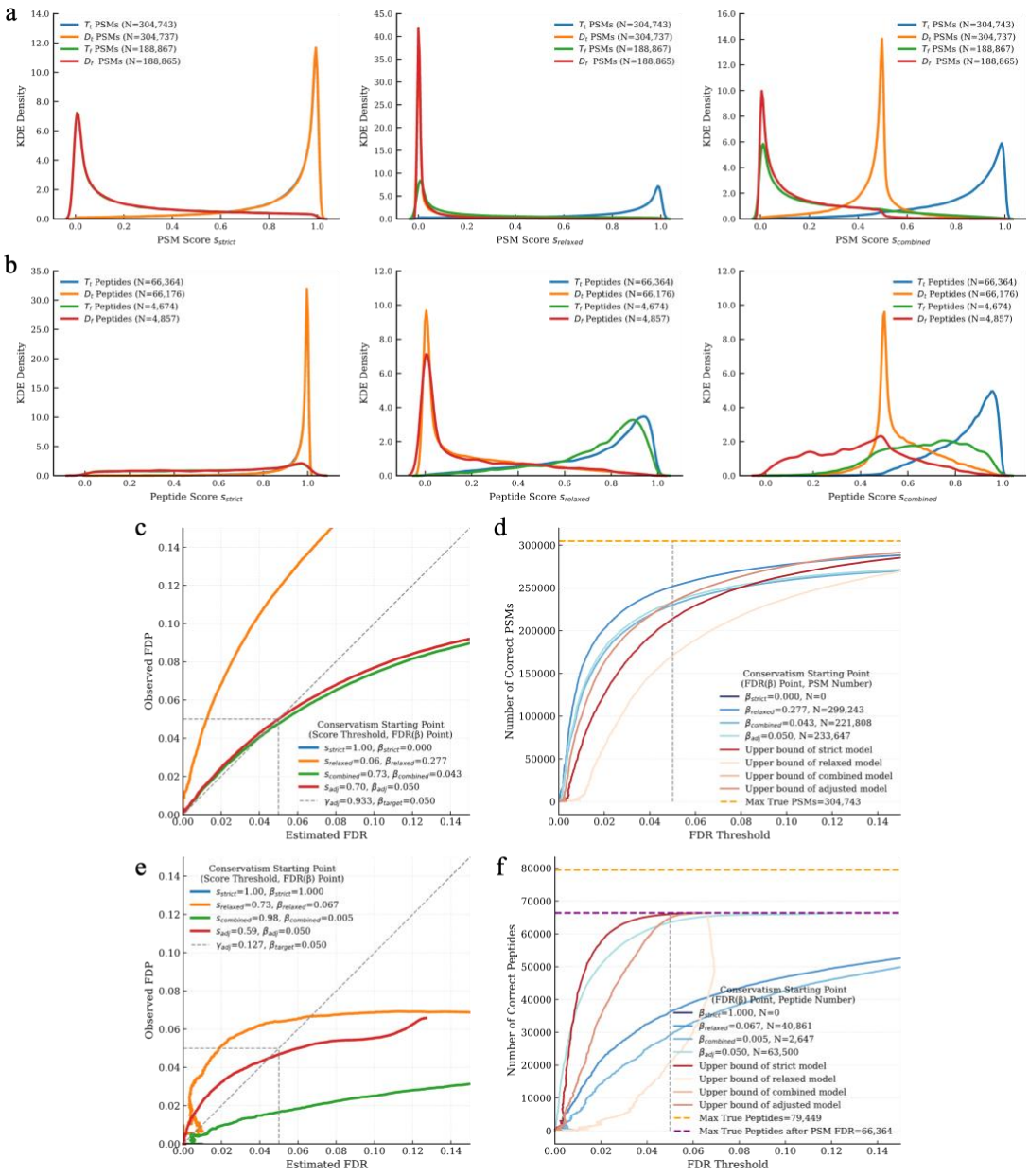


Figure S52: Dual-level CCT QC on replicate 2 of Astral DIA dataset (3p5ms) using π-xNovo-DIA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.

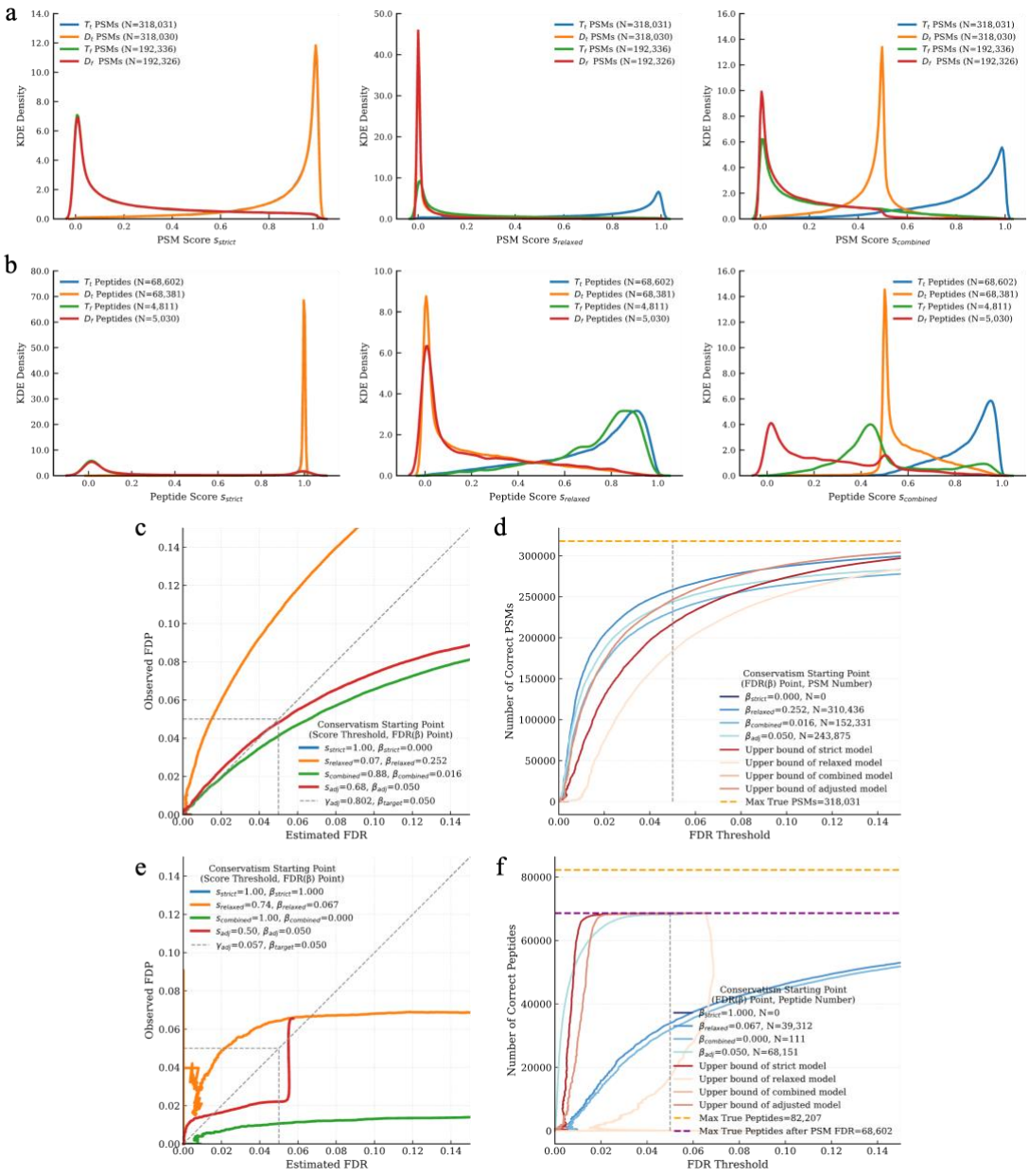


Figure S53: Dual-level CCT QC on replicate 3 of Astral DIA dataset (3p5ms) using π-xNovo-DIA. **a, b**, PSM level (a) and peptide level (b)**:** score density distributions of the four match categories (T_t, T_f, D_t, D_f) under the strict, relaxed and combined scoring models. **c, e,** PSM level (c) and peptide level (e): relationship between the estimated FDR (β) and the observed false discovery proportion (FDP) across score thresholds; the dashed line denotes the identity (y=x). **d, f,** PSM level (d) and peptide level (f): numbers of retained true-positive matches as a function of the FDR threshold for each model; dashed lines indicate (i) the theoretical upper bound of true-positive peptides and (ii) the upper bound after PSM-level FDR filtering.